SANOS-Evolve: A Discrete Stochastic-Local-Volatility Model for European-Option Smile Dynamics*

Shaosai Huang
Kspectra Research Inc., Toronto, ON, M2N 0G3, Canada
`arthur.foxie.huang@kspectra.ai`

July 20, 2026

Abstract

An arbitrage-free set of option marginals fixes an underlying's local-volatility content, but not how its smile moves when the spot moves. Existing models are either parametric in the smile (stochastic volatility) or exact but expensive to calibrate (stochastic-local volatility). SANOS-Evolve keeps the two separate: a static SANOS linear program fixes strictly arbitrage-free marginals, and a dynamic layer fits a martingale two-timescale kernel between them, calibrated while the marginals stay fixed and arbitrage-free by construction (Strassen). Every object is a Gaussian mixture, so prices, Greeks, and the variance-index smile are closed-form. Translation invariance makes the skew-stickiness ratio an exact closed-form realised covariance whose term structure the two timescales fit, and identifies the vol-of-vol from a forward-variance readout—VIX for SPX, the strip's own curvature otherwise. One calibrated object yields the surface, exotics, and SSR-consistent hedges; on SPX the delta reproduces the industry Hull–White minimum-variance benchmark from the smile alone, with no trailing realised estimate.

Contents

*Working paper. Comments welcome.

1 Introduction

An equity-index volatility surface poses two distinct problems. The *static* problem is to fit, at each listed maturity, a risk-neutral marginal that reprices the quoted vanillas within bid–ask and is free of butterfly and calendar arbitrage. The *dynamic* problem is to say how that surface moves when the spot moves. That smile dynamics governs forward-starting products, path-dependent exotics, and hedge ratios. Solving the static problem does not solve it.

Statics: marginals are local volatility. By Dupire's formula [2], a complete set of arbitrage-free marginals across maturities *is* the local-volatility content of the surface. The local variance $\sigma^2_{\mathrm{LV}}(K,T) = \partial_T C/(\frac{1}{2}K^2\partial_{KK}C)$ depends only on the marginals. SANOS [1] solves this static problem: it writes call prices as weighted sums of Black–Scholes basis functions and fits the weights by a linear program (LP) with hard bid–ask and calendar constraints. The output is a smooth, strictly arbitrage-free marginal at each maturity. It also induces a discrete Markov martingale whose canonical transition operator is of the rigid local-volatility type.

Dynamics: three families, and a gap. The local-volatility model implied by those marginals [4, 5], however, has the *wrong* dynamics. Its forward smiles flatten too quickly with maturity. And its skew-stickiness ratio—the SSR, the normalised co-movement of at-the-money volatility with spot (Section 6)—is a rigid consequence of the static skew, not a free target. Stationary local volatility gives the linear limit $\mathrm{SSR} \to 2$; calibrated to the empirical $\tau^{-1/2}$ skew it is pinned near the Doeff–Kamal value $\mathrm{SSR} \approx \frac{3}{2}$ [33]. The empirical SPX term structure instead runs 1.9, 1.45, 1.0 from one week to two years [22, 23].

Stochastic volatility puts the randomness in the volatility itself and produces genuine smile dynamics. But it pays twice. Its parametric marginals cannot reprice the vanilla smile exactly: Heston [9] and SABR [10] miss the short-dated SPX skew. Its dynamics are rigid too. Calibrated to the same smile, the whole family—Heston, two-factor and rough Bergomi [23]—clusters on one SSR term structure, and that cluster deviates from the empirical one [27, 25].

Stochastic-local volatility (SLV) restores the exact smile by multiplying a stochastic backbone by a local-volatility leverage, calibrated through Gyöngy's theorem [6, 7, 8]. The cost is that the leverage is a fixed point, solved by particle or PDE methods [15, 14], more recently by optimal transport [16] or neural SDEs [17]. So there is no closed-form transition kernel, and the Greeks carry Monte-Carlo noise. And because the leverage is spent matching the smile, the SSR is left to the backbone and inherits its rigidity (Section 8 makes the comparison precise).

So the field offers three incomplete options: exact but static (local volatility), dynamic but parametric (stochastic volatility), and exact and dynamic but intractable (SLV). Missing is a kernel that is at once tractable (closed-form), data-driven (fits any arbitrage-free smile exactly), and dynamically controllable (the SSR a free target). SANOS-Evolve is that kernel.

SANOS-Evolve: marginals plus a fitted kernel. SANOS supplies arbitrage-free marginals $\{\mu_j\}$, but pairs them with a rigid canonical transition operator; its authors leave controllable discrete-time dynamics as an open direction [1]. SANOS-Evolve keeps the marginals and replaces that operator. In its place we fit a martingale transition kernel $\mathcal{K}_j$ between consecutive marginals, carrying a latent volatility regime with two mean-reversion timescales:

$$\underbrace{\{\mu_j\}}_{\substack{\text{SANOS marginals}\\ \text{(data-driven statics)}}} \quad + \quad \underbrace{\{\mathcal{K}_j\}}_{\substack{\text{martingale kernel}\\ \text{(controllable dynamics)}}} \quad = \underbrace{\text{closed-form discrete SLV}}_{\text{for any European option.}}$$

We fix the marginals from quotes and fit the kernel between them, rather than postulate a process and back out a fit. The SANOS no-arbitrage structure is therefore preserved exactly while the dynamics are calibrated. Two properties make this work.

The kernel is expressive enough. A Gaussian mixture placed in the *kernel*—not, as in Brigo–Mercurio [13], in the marginal of a shared-forward diffusion—is dense, in Wasserstein-1, among the *translation-invariant* dynamics a kernel can carry: any SSR term structure, any vol-of-vol (Appendix C). Matching the marginals is *not* the kernel's job—a deterministic leverage overlay supplies it (below)—so the marginal fit is never the binding constraint. We keep a parsimonious two-timescale form, which pins the dynamics to identifiable regimes rather than leaving them underdetermined.

We know what the kernel must fit. The smile's motion, as distinct from its shape, decomposes into a small basis, graded by the order each term takes in the vol-of-vol expansion [23]: a *leverage* (spot–vol coupling, first order), a *vol-of-vol* (second order), and a *vol-of-vol skew* (third order). Each is carried across maturity by the volatility process's timescales, and at the short end by its roughness. The SSR is the leading coordinate: the leverage normalised by the static skew, $\mathrm{SSR}_T = -\beta_T\sqrt{T}/\mathcal{S}_T$ with $\beta_T = \mathrm{d}\widehat{\sigma}_{\mathrm{ATM}}/\,\mathrm{d}\log S$. Fitting it therefore decouples the dynamic response from the static shape. SANOS-Evolve calibrates the first- and second-order dynamics (the SSR, a VIX-informed vol-of-vol readout, and the two timescales) separately from the statics; higher-order terms are named extensions (Section 6). The variance index enters as information rather than a target: a soft penalty that *identifies* the kernel's vol-of-vol. This is what separates the construction from joint SPX/VIX calibration: dispersion-constrained bridges [31] and the perturbed-OT coupling [32] both fit the two smiles as hard targets. A hard VIX target presumes a volatility-index *options* market deep enough to calibrate across strikes. Among European underlyings that effectively means SPX alone: VSTOXX and VXN options exist but trade an order of magnitude thinner, and no single name, FX pair, or crypto has a VIX-comparable one. A forward-variance *readout* needs only the index level or futures where they exist, or the option strip's own curvature otherwise. It therefore identifies the vol-of-vol for far more underlyings (Table 2). Reading it is the general construction, of which the SPX+VIX case is the sharpest instance. The price is honest: more portable, but less tightly identified precisely where a volatility index does exist (Section 6). The basis, with the kernel knobs (Section 4) that carry each characteristic, is:

characteristic	order	smile signature	in the kernel
leverage (the SSR)	1st	skew + its stickiness	$\lambda_{\mathrm{skew}}, \lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$
vol-of-vol	2nd	convexity, forward-smile spread, VIX level	$\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}$
vol-of-vol skew	3rd	VIX-smile slope, deep-wing asymmetry	— (extension)
timescales	—	maturity-decay of the above	$\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$
roughness H	—	short-T blow-up of skew/SSR	$H=\frac{1}{2}$ fixed (extension)

Figure 1 places SANOS-Evolve among transition-kernel families by three properties: the explicitness of the kernel (horizontal), the dynamic freedom (vertical), and the exactness of the marginal fit (fill).

Contributions.

1. **A discrete SLV that decouples statics from dynamics (Sections 2–3).** A convex LP fixes strictly arbitrage-free marginals; a separately fitted martingale kernel carries the dynamics, and the marginals stay invariant when the dynamic targets are re-tuned. Any admissible chain is free of static arbitrage, with existence guaranteed by convex order (Strassen [20]). There is no leverage fixed point and no existence cliff, unlike classical SLV.

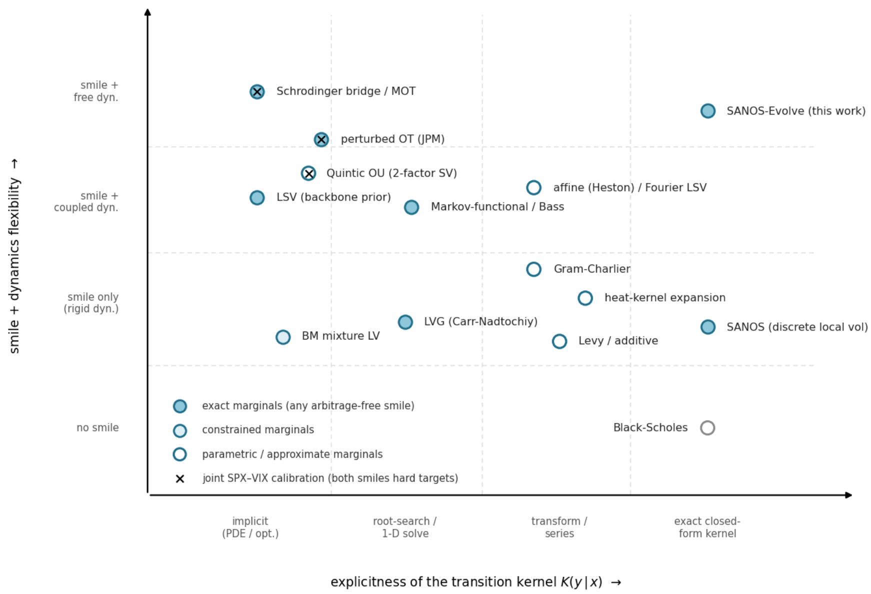

Figure 1: **The transition-kernel landscape.** Explicitness of the one-step kernel $K(y \mid x)$ (horizontal), smile-and-dynamics freedom (vertical), exactness of the marginal fit (fill); tiers and legend on the plot (Black–Scholes: no smile, fit n/a, grey). In the rightmost column, SANOS-Evolve replaces SANOS's rigid canonical coupling with a free-dynamics Gaussian-mixture kernel (Appendix C). A black cross marks joint SPX–VIX calibration (both smiles as hard targets)—the programme the VIX-as-readout lane of Section 1 stays out of. *Abbreviations:* MOT/OT, (martingale) optimal transport; LSV, stochastic-local volatility; LVG, local variance gamma; BM, Brigo–Mercurio; LV, local volatility; SV, stochastic volatility; OU, Ornstein–Uhlenbeck.

2. **The SSR as an exact closed-form realised covariance, free of the smile (Sections 4, 6).** A translation-invariant two-timescale kernel places the spot–vol leverage in the return innovation around fast and slow regimes. Translation invariance then makes the SSR an exact realised covariance of return and regime—not a spot-derivative or a soft penalty—and the two timescales give its term structure, fit separately from the smile. This is the freedom the calibrated stochastic-volatility family lacks.

3. **Closed-form and deterministic (Sections 5, 6).** Prices, Greeks, the variance-index smile, and the SSR and forward-variance readouts are finite Black–Scholes mixtures of the translation-invariant kernel; the leverage-fused propagation that restores the marginals adds only a first-order collocation, held within bid–ask (Section 4.6). Hedge ratios are therefore reproducible and noise-free, where particle SLV carries Monte-Carlo noise and PDE SLV carries grid error. The vol-of-vol is identified from a variance-index (VIX) soft target rather than by jointly calibrating a volatility-index market.

Related work. The closest dynamic relative is regime switching: Jourdain and Zhou [18] prove existence of a calibrated regime-switching model, a special case of local-stochastic volatility in which a finite-state process, not the spot, carries the randomness. Our latent regime is finite-state in the same spirit, but two-timescale and calibrated to the SSR term structure. The fast/slow structure itself is standard multi-factor stochastic volatility in the Bergomi lineage [22]. The Quintic OU model [28]—the same two-OU state under a polynomial volatility map—likewise steers the SSR separately: even its joint SPX/VIX calibration needs an SSR band penalty, independent evidence that the smiles do not pin the SSR. Its polynomial map buys one closed-form readout, the VIX functional; we keep the exponential map and discretise the *state* (Section 4), which buys the whole kernel. That forward implied volatility needs a transition density, not merely marginals, is the lesson of [19].

We build on martingale optimal transport [20, 21]: we adopt its admissible-set constraint (with Strassen existence) but replace its objective, fitting an economically structured kernel to dynamic targets rather than extremising an exotic's price. A neural mixture-density surrogate [30] reuses the same Gaussian-mixture and CDF-loss machinery for the opposite, simulation-surrogate objective.

Scope. SANOS-Evolve applies to any European-option underlying. The construction needs only European exercise (so the terminal marginal is read from prices), a liquid surface, and a forward-variance readout for the vol-of-vol. We take SPX as the worked instance, where a liquid VIX gives the sharpest readout; equity indices follow directly, and FX, crypto, and European single names reuse the same machinery with their own readout and SSR (Section 6). Throughout, we use a readout to *identify* vol-of-vol, not to jointly calibrate a volatility-index options market—the route of the perturbed-OT desk method [32]. Generality here is structural, a property of the construction rather than an empirical claim across asset classes: the study is SPX. The daily framework is a methods contribution with ground-truth validation and a real-data calibration (Section 7); a brief intraday/0DTE outlook closes the discussion (Section 8).

2 The static layer: SANOS marginals as local volatility

Fix a pricing date and maturities $T_0 < T_1 < \cdots < T_M$ with forwards F_j and discount factors D_j. Work in log-moneyness $z = \log(S/F)$. SANOS fits, at each T_j, an arbitrage-free density as a Gaussian mixture in z,

$$\rho_{S,T_j}(z) = \sum_i q_j^i\, \mathcal{N}\big(z; \mu_j^i, (\sigma_j^i)^2\big), \qquad q_j^i \geq 0,\ \sum_i q_j^i = 1, \tag{1}$$

by a linear program over the weights q_j subject to hard bid–ask boxes, forward (martingale) normalisation, and calendar constraints [1]. We denote the resulting marginal law $\mu_j = \mathrm{Law}(S_{T_j})$. By Breeden–Litzenberger [3], (1) is butterfly-arbitrage-free iff $\rho_{S,T_j} \geq 0$, which holds by construction. The family $\{\mu_j\}$ is calendar-arbitrage-free iff it is increasing in convex order, which the SANOS LP enforces. By Dupire [2], $\{\mu_j\}$ *is* the local-volatility content of the surface.

Two structural facts make the whole construction closed-form. First, in log-coordinates every object we use is a Gaussian mixture. Second, Black–Scholes is not a modelling assumption but a computational identity: the call price under a single Gaussian log-return is the moment-generating function of a truncated Gaussian,

$$\mathbb{E}\big[(S-K)^+\big] = F\,\Phi(d_1) - K\,\Phi(d_2), \qquad d_{1,2} = \frac{\log(F/K) \pm \frac{1}{2}\tau^2}{\tau}, \tag{2}$$

with $\tau = \sigma\sqrt{T_j}$ the total volatility. The price of any claim is then a *pairing* of the marginal with its payoff; for the mixture (1) the call is a finite weighted sum of Black–Scholes evaluations,

$$C_j(K) = \big\langle \mu_j, (\cdot - K)^+ \big\rangle = \sum_i q_j^i\ \mathrm{BS}\big(F_j^i, K, \sigma_j^i\big), \qquad F_j^i = F_j\, e^{\mu_j^i + \frac{1}{2}(\sigma_j^i)^2}, \tag{3}$$

with no quadrature, no PDE, and no Monte Carlo. The same holds for digitals, and for any payoff with a closed-form Gaussian expectation. Because call payoffs separate measures (Breeden–Litzenberger), this pairing is a faithful coordinate for μ_j: the mixture density makes the marginal matchable, and the closed-form pairing makes it readable.

The same mixture sum gives the Greeks in closed form—each a q-weighted sum of component Black–Scholes sensitivities, no bumping and no Monte-Carlo noise. With $d_1^i = [\log(F_j^i/K) + \frac{1}{2}(\sigma_j^i)^2]/\sigma_j^i$,

$$\Delta = \frac{1}{S}\sum_i q_j^i F_j^i\, \Phi(d_1^i), \qquad \mathcal{V} = \sum_i q_j^i F_j^i\, \varphi(d_1^i), \qquad \Gamma = \frac{1}{S^2}\sum_i \frac{q_j^i F_j^i\, \varphi(d_1^i)}{\sigma_j^i}, \tag{4}$$

with Δ, Γ in spot and $\mathcal{V}$ per unit total volatility ($\times\sqrt{T_j}$ for the annualised vega); higher sensitivities follow the same way. These are the *sticky-moneyness* Greeks of the fixed marginal, computed with the smile held fixed in moneyness (SSR $= 0$). The desk hedge delta instead rolls the smile at the calibrated SSR—the SSR-consistent Δ_{SSR} of (26).

3 The dynamic layer: admissible transition kernels

With the marginals fixed, this section defines what may move between them. We set up the Markov kernel on the joint spot–regime state, then carve out the *admissible* set: positive, normalised, martingale, marginal-matching. On that set, absence of static arbitrage is a theorem and existence is Strassen's. Every dynamic choice in the paper is made *within* this set; Section 4 selects the element.

3.1 State and kernel

The dynamic object is a Markov chain on the joint state

$$X_j = (Z_j, f_j, s_j), \qquad Z_j = \log(S_{T_j}/F_j), \quad f_j \in \{1,\dots,n_{\mathsf{F}}\}, \quad s_j \in \{1,\dots,n_{\mathsf{S}}\}, \tag{5}$$

where Z_j is log-spot and (f_j, s_j) are a *fast* and a *slow* latent volatility regime—*both carried* across maturities, the genuine two-timescale (Bergomi) memory. The transition over $[T_j, T_{j+1}]$ is a Gaussian-mixture kernel. It places the spot–vol leverage in the return *innovation*—a within-step node $l = 1,\dots,n_{\mathsf{L}}$ correlated with the regime moves—around a *fixed* stationary vol target:

$$\mathcal{K}\big((z,f,s),(\,\mathrm{d}z',f',s')\big) = \sum_{l=1}^{n_{\mathsf{L}}} w_l\, \underbrace{P_{\mathsf{F}}(f' \mid l, f)\, P_{\mathsf{S}}(s' \mid l, s)}_{\text{two carried OU factors}}\, \mathcal{N}\big(\,\mathrm{d}z';\, z + d_{f,s,l},\, V_{f,s,l}\big), \tag{6}$$

with $\mathcal{N}(\,\mathrm{d}z'; m, V)$ the Gaussian measure with mean m and variance V. The *coefficients* on the right—the weights w_l, the transitions $P_{\mathsf{F}}, P_{\mathsf{S}}$, the drift $d_{f,s,l}$ and variance $V_{f,s,l}$—are functions of the regime indices alone. (Throughout, sans-serif $\mathsf{F}, \mathsf{S}, \mathsf{L}$ name the three factors—fast regime, slow regime, within-step return—the matching italic f, s, l are their running node indices, and $n_{\mathsf{F}}, n_{\mathsf{S}}, n_{\mathsf{L}}$ the node counts; the factors' Ornstein–Uhlenbeck structure is Section 4.) The spot z enters *only* as the additive centre of the Gaussian, so the kernel depends on (z, z') solely through the return $z' - z$. It is therefore **translation-invariant in log-spot**: the conditional return law—and hence the smile in moneyness—depends on the regime (f, s) alone, not on the spot level. This single property keeps every dynamic readout (the SSR in particular) an exact closed form (Section 6) and makes propagation an exact Gaussian-mixture map (Section 4). The within-step node l carries the instantaneous skew and is renewed each step; the two regimes are carried across steps.

Spot marginal vs. lifted law. The kernel acts on the joint state, so its one-step law is a *lifted* joint $\widehat{\mu}_j(\mathrm{d}z, f, s)$, whereas SANOS supplies only the spot marginal $\mu_j^Z(\mathrm{d}z) = \sum_{f,s} \widehat{\mu}_j(\mathrm{d}z, f, s)$. The first maturity is lifted with the stationary joint regime law $\pi_{f,s}$; thereafter the joint is carried by the propagation. The shorthand $\mu_j \mathcal{K}_j = \mu_{j+1}$ denotes the match of *spot* marginals (Section 3.3).

3.2 Admissibility and no static arbitrage

Definition 1 (Admissible kernel)**.** *Given forward-normalised marginals μ_j, μ_{j+1}, a Markov kernel $\mathcal{K}_j$ is* admissible *if*

$$\mathcal{A}_j = \Big\{ \mathcal{K}_j : \ \mathcal{K}_j \geq 0, \ \mathcal{K}_j \mathbf{1} = 1, \ \mathbb{E}[e^{Z_{j+1}} \mid X_j = x] = e^{Z_j} \ \forall x, \ \mu_j \mathcal{K}_j = \mu_{j+1} \Big\}. \tag{7}$$

The conditions are, in order: positivity, normalisation, the (forward-normalised) martingale property, and exact marginal propagation.

Theorem 1 (No static arbitrage)**.** *Let $\{\mu_j\}_{j=0}^M$ be probability measures on $(0, \infty)$ with $\mathbb{E}_{\mu_j}[S] = F_j$. Assume their forward-normalised laws $\tilde{\mu}_j = \mathrm{Law}(S_{T_j}/F_j)$ are increasing in convex order, $\tilde{\mu}_0 \preceq_{\mathrm{cx}} \tilde{\mu}_1 \preceq_{\mathrm{cx}} \cdots \preceq_{\mathrm{cx}} \tilde{\mu}_M$. If $\mathcal{K}_0, \ldots, \mathcal{K}_{M-1}$ are admissible (each $\mathcal{K}_j \in \mathcal{A}_j$), then the discrete-time process with initial law μ_0 and transitions $\{\mathcal{K}_j\}$ is a martingale after discounting, with one-dimensional marginals exactly $\{\mu_j\}$. The induced call surface $C_j(K) = D_j \int (s-K)^+ \mu_j(\mathrm{d}s)$ is then free of strike (butterfly) and calendar static arbitrage. Moreover $\mathcal{A}_j \neq \emptyset$ for every j.*

A proof sketch is given in Appendix B. The content is that the no-arbitrage *claim attaches to the admissible set*, and that the set is nonempty precisely because SANOS delivers marginals in convex order (Strassen). SANOS's own canonical discrete-local-volatility coupling [1] is one element of $\mathcal{A}_j$: the rigid one the introduction set out to replace. The dynamic targets of Sections 5–6 (SSR, VIX, forward starts) *select a different, controllable element within $\mathcal{A}_j$*. They never replace the set.

3.3 The statics/dynamics decoupling

The kernel's translation invariance carries a structural price. Because it depends on (z, z') only through the return, $\mu_j \mathcal{K}_j$ is a pure convolution: it adds the same increment law at every spot and cannot reshape the smile, so a translation-invariant kernel cannot on its own meet the marginal-matching condition of (7). Matching is restored by one deterministic ingredient, a *leverage overlay* $\sigma_{\mathrm{LV}}(z, T_j)$—the discrete stochastic-local-volatility structure—which Gyöngy's identity fixes so the overlaid kernel lands on the SANOS target *by construction* (formula (17), Section 4.6). This closes admissibility *and decouples* the two layers: the leverage carries the statics, the kernel the dynamics (the SSR). The first three admissibility conditions are then structural in the coefficient map, the fourth is the leverage, and calibration fits only the *dynamic* knobs while the statics come along for free—no fit-then-project. The division of labour is also what keeps the kernel's task achievable: it need only be flexible in the *dynamics*, where the translation-invariant mixture kernel is dense (Appendix C), and never match marginals, which no amount of kernel richness alone could do. This decoupling is the structural core of the construction; Section 4.6 makes the overlay quantitative.

4 The two-timescale latent-regime kernel

This section constructs the element of $\mathcal{A}_j$ we use: the two-factor Bergomi latent variance, Gauss–Hermite-discretised into a finite regime chain driven by nine structural knobs, with the spot–vol leverage placed in the return innovation. Three ingredients are arranged to be exact: the per-fibre martingale lock, the Gaussian-mixture closure under propagation, and the Gyöngy leverage that lands the propagated marginals on the SANOS targets. Recompression and semigroup interpolation then complete the toolkit.

4.1 Two carried timescales and the Gauss–Hermite discretisation

The latent log-variance is a baseline plus two mean-reverting Ornstein–Uhlenbeck factors: a *fast* factor X^{F} (rate κ_{F}) and a *slow* factor X^{S} (rate κ_{S}). So $x_j = \bar{\gamma} + X_j^{\mathsf{F}} + X_j^{\mathsf{S}}$, the continuous two-factor Bergomi model. Conditional on the latent vol the one-step log-return is Gaussian. But its mean and variance are *functions of the continuous vol state*, so the kernel is a continuous, variable-coefficient mixture of Gaussians: an integral over the latent factors, *not* a finite Gaussian mixture, and not closed form. The remedy is quadrature. An n-point **Gauss–Hermite** rule represents $\mathcal{N}(0,1)$ by n nodes ζ_l and weights w_l reproducing the Gaussian's moments through order $2n-1$,

$$\mathbb{E}_{\mathcal{N}(0,1)}[g] = \int g(\zeta)\,\frac{e^{-\zeta^2/2}}{\sqrt{2\pi}}\,\mathrm{d}\zeta \;\approx\; \sum_{l=1}^{n} w_l\, g(\zeta_l) \qquad (\text{exact for } \deg g \le 2n-1), \tag{8}$$

the moment-optimal finite stand-in for a Gaussian. Replacing each continuous latent quantity by its n GH nodes turns the integral into a finite sum, each node a *vol scenario*. The kernel then becomes a finite Gaussian mixture over a quadrature grid. *Both* vol factors are carried:

- **Fast factor** F—carried regime, node index $f = 1, \ldots, n_{\mathsf{F}}$, mean-reversion κ_{F} (the high, decaying short end of the SSR);
- **Slow factor** S—carried regime, node index $s = 1, \ldots, n_{\mathsf{S}}$, mean-reversion κ_{S} (the sustained long-end floor);
- **Within-step return** L—node index $l = 1, \ldots, n_{\mathsf{L}}$, *renewed* each step, carrying the instantaneous skew and the innovation leverage.

The nodes $\{\zeta_f^{\mathsf{F}}\}, \{\zeta_s^{\mathsf{S}}\}, \{\zeta_l^{\mathsf{L}}\}$, base log-weights $\eta^{\mathsf{F}}, \eta^{\mathsf{S}}$, and within-step weights $w_l = \mathrm{softmax}(\eta^{\mathsf{L}})$ are *fixed* quadrature constants; the data fit only the nine structural knobs below. The vol-of-vols $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}$ scale the nodes, the leverages $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ couple the regime moves to the realised return node l (never to z), and λ_{skew} tilts the within-step return. The node counts are a *modelling* choice, not an accuracy dial: the within-step factor converges at $n_{\mathsf{L}} = 5$, but the carried factors converge slowly ($V \propto e^{\nu\zeta}$ is lognormal in the node, so Gauss–Hermite resolves $\mathbb{E}[V]$ only gradually at large $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$—the ATM level still drifts at $n_{\mathsf{F}} = 9, n_{\mathsf{S}} = 7$). We keep a coarse $n_{\mathsf{F}} = 5, n_{\mathsf{S}} = 3$ for pricing and the SSR and let calibration absorb the discretisation: with $\bar{\gamma}$ and the leverages fit *at* the grid, the model hits its ATM and SSR targets at any count and refining merely re-shifts $\boldsymbol{\theta}$. The vol-of-vol is the one exception—a high-moment functional that converges only near n_{F}=17, where a coarse count spuriously inflates it—so the cross-regime validation of Section 7.2 both calibrates *and* reads it at n_{F}=17 (fit and readout on the *same* converged grid), while the low-moment deliverables keep the production count. The continuous limit is thus a structural relationship, not a convergence claim at these counts; the counts trade resolution against identifiability (Appendix D) and cost.

4.2 The coefficient map

The component coefficients are *determined* by nine structural knobs $\boldsymbol{\theta} = (\bar{\gamma}, \nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}, \lambda_{\text{skew}}, \lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}, \kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$. The per-node variance and the within-step skew tilt are

$$V_{f,s,l} = \exp\!\big(\bar{\gamma} + \nu_{\mathsf{F}}\,\zeta_f^{\mathsf{F}} + \nu_{\mathsf{S}}\,\zeta_s^{\mathsf{S}} + \nu_{\mathsf{L}}\,\zeta_l^{\mathsf{L}}\big)\Delta_j, \qquad (\text{two-factor variance}), \tag{9}$$

$$\ell_{f,s,l} = \lambda_{\text{skew}}\,\zeta_l^{\mathsf{L}}\,\sqrt{V_{f,s,l}}, \qquad (\text{instantaneous mean-tilt / skew}), \tag{10}$$

and the two carried factors evolve by leverage-coupled mean-reverting transitions whose tilt is driven by the realised return node l, *not* by log-spot:

$$P_{\mathsf{F}}(f' \mid l, f) = \operatorname{softmax}_{f'}\big(\eta^{\mathsf{F}}_{f'} + \kappa_{\mathsf{F}}\,\zeta^{\mathsf{F}}_{f}\zeta^{\mathsf{F}}_{f'} + \lambda_{\mathsf{F}}\,\zeta^{\mathsf{L}}_{l}\,\zeta^{\mathsf{F}}_{f'}\big), \tag{11}$$

$$P_{\mathsf{S}}(s' \mid l, s) = \operatorname{softmax}_{s'}\big(\eta^{\mathsf{S}}_{s'} + \kappa_{\mathsf{S}}\,\zeta^{\mathsf{S}}_{s}\zeta^{\mathsf{S}}_{s'} + \lambda_{\mathsf{S}}\,\zeta^{\mathsf{L}}_{l}\,\zeta^{\mathsf{S}}_{s'}\big). \tag{12}$$

Here η pulls to the centre (mean reversion), κ to the current regime (persistence / timescale), and $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ are the *innovation* leverages. A down move (ζ^{L}_l large via $\lambda_{\text{skew}} < 0$) tilts the regimes to higher vol. Because the leverage multiplies the innovation around a *fixed* stationary target, the equilibrium vol is *not* a deterministic function of spot. It is genuine stochastic volatility, and the kernel stays translation-invariant in z.

4.3 Exact per-fibre martingale normalisation

The within-step drift is re-locked to the forward on *every* fibre—the kernel's conditional law at a fixed state (z, f, s). With raw mean $\widetilde{m}_{f,s,l} = -\frac{1}{2}V_{f,s,l} + \ell_{f,s,l}$, subtract the per-regime constant

$$A_{f,s} = \log \sum_l w_l \, \exp\big(\widetilde{m}_{f,s,l} + \tfrac{1}{2}V_{f,s,l}\big) = \log \sum_l w_l \, e^{\ell_{f,s,l}}, \qquad d_{f,s,l} = \widetilde{m}_{f,s,l} - A_{f,s}, \tag{13}$$

so that $\mathbb{E}[e^{Z_{j+1}} \mid z, f, s] = e^z$ exactly. The shift is a single per-fibre constant, *independent of* z (the within-step weights w_l are the fixed Gauss–Hermite weights). It is stable and differentiable, and it preserves the relative branch tilts that carry the leverage and the SSR. It replaces any penalty-only enforcement and the fragile "adjust one component mean" construction, which fails when the required numerator turns non-positive. The kernel is now complete: a finite Gaussian-mixture martingale on (z, f, s).

4.4 The SSR dial

The kernel is translation-invariant (Section 3), so the conditional smile $\sigma(T; f, s)$ depends only on the regime pair. The skew-stickiness ratio is then generated by the *innovation* leverage $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ (the return$\leftrightarrow$regime correlation), not by any z-coupling: $\lambda_{\mathsf{F}} = \lambda_{\mathsf{S}} = 0 \Rightarrow$ no spot–vol comovement $\Rightarrow$ SSR $= 0$ (sticky-moneyness). The dials and their term-structure shaping are developed in Section 6. The SSR is thus not a spot-derivative of a state-dependent smile but the closed-form return $\times$ regime covariance of Section 6. The placement of the leverage is a design decision with teeth. A Monte-Carlo cross-check showed that coupling the regimes to *spot* instead makes the equilibrium vol a deterministic function of spot (local volatility). That pins SSR $\to 2$ at long maturities and cannot be re-tuned. Placing the leverage in the innovation breaks that floor while keeping the readouts exact.

4.5 Exact Gaussian-mixture closure

The point of translation invariance is that one propagation step of the discretised kernel is an *exact Gaussian convolution*: the finite mixture of Section 4 stays a finite mixture, so every downstream readout is a closed-form sum rather than a quadrature. The variable coefficients of Section 4 do not obstruct this, because their variability is over the *vol state*, not the spot: Gauss–Hermite has already replaced the continuous vol state by the *discrete* regime index, so $d_{f,s,l}, V_{f,s,l}$ take finitely many values (one per regime), and translation invariance (Section 3) makes each independent of the spot z.

Conditioned on a regime, then, the kernel adds a *fixed* Gaussian increment $\Delta z \sim \mathcal{N}(d_{f,s,l}, V_{f,s,l})$—constant in the integration variable—and fixed Gaussians convolve in closed form, means and variances adding:

$$\mathcal{N}(\,\cdot\,; m_c^{f,s}, v_c^{f,s}) \;*\; \mathcal{N}(\,\cdot\,; d_{f,s,l}, V_{f,s,l}) \;=\; \mathcal{N}\big(\,\cdot\,;\, m_c^{f,s} + d_{f,s,l},\; v_c^{f,s} + V_{f,s,l}\big). \tag{14}$$

So a law carried, per regime (f, s), as a Gaussian mixture in log-price $\sum_c \omega_c^{f,s} \mathcal{N}(m_c^{f,s}, v_c^{f,s})$ propagates to another finite mixture,

$$\rho_{j+1}(z', f', s') = \sum_{f,s,\,c} \omega_c^{f,s} \sum_l w_l\, P_{\mathsf{F}}(f' \,|\, l, f)\, P_{\mathsf{S}}(s' \,|\, l, s)\, \mathcal{N}\big(z';\, m_c^{f,s} + d_{f,s,l},\; v_c^{f,s} + V_{f,s,l}\big), \tag{15}$$

each new component the convolved Gaussian of (14), reweighted by the within-step weight w_l and the regime transitions. The coefficient variability lives entirely in this outer sum over (f, s, l), never inside a convolution, so GM $*$ GM $=$ GM holds exactly. No quadrature, PDE, or Monte Carlo appears anywhere in the chain. (A *state-dependent* kernel—z-dependent weights $w_l(z)$ or variances—breaks (14). Propagation then relapses to a continuous integral mixture, forcing a first-order collocation that a Monte-Carlo cross-check showed biases the multi-step SSR.) The lone cost is that (15) multiplies the component count by $n_{\mathsf{L}} n_{\mathsf{F}} n_{\mathsf{S}}$ each step. The recompression of §4.7 below holds it to a fixed budget by a moment-preserving, martingale-relocked reduction. Convex-order / calendar no-arbitrage rests on the martingale kernel, not on this exact finite closure.

4.6 The discrete SLV fusion: leverage carries the statics

The leverage overlay of the decoupling (Section 3.3) is made quantitative here. Split the increment into a local scale and a unit stochastic-vol factor,

$$\Delta z \;\sim\; \bar d_{f,s} + \underbrace{\sigma_{\mathrm{LV}}(z, T_j)\sqrt{\Delta_j}}_{\text{statics (SANOS)}} \cdot \underbrace{\sqrt{\nu_{f,s}}}_{\text{SSR (kernel)}}\;\xi, \qquad \nu_{f,s} = \frac{\bar V_{f,s}}{\mathbb{E}_\pi[\bar V]}, \quad \bar V_{f,s} = \mathbb{E}_l[V_{f,s,l}] + \mathrm{Var}_l(d_{f,s,l}), \tag{16}$$

where ξ is a standard (unit-variance) innovation, $\bar d_{f,s}$ the per-fibre martingale-lock drift (13), and $\bar V_{f,s}$ the kernel's *full* one-step increment variance by the law of total variance: the within-node $\mathbb{E}_l[V_{f,s,l}]$ plus the between-node mean spread $\mathrm{Var}_l(d_{f,s,l})$ that the skew tilt λ_{skew} creates. Using $\mathbb{E}_l[V]$ alone mis-scales the leverage by $\sim 3\times$ at the calibrated λ_{skew}. The unit factor $\nu_{f,s}$ ($\mathbb{E}_\pi[\nu] = 1$, with italic regime indices—not the sans-labelled vol-of-vol amplitudes $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}$) keeps the kernel's fluctuation, the regime comovement, i.e. the SSR. It drops the level, and σ_{LV} sets that level from the market. Marginal-matching is then Gyöngy's identity [6] in discrete form,

$$\sigma^2_{\mathrm{LV}}(z, T_j) = \frac{\sigma^2_{\mathrm{Dupire}}(z, T_j)}{\mathbb{E}[\nu \mid z]}, \tag{17}$$

with σ_{Dupire} from the SANOS marginals (closed-form Gaussian-mixture call + Breeden–Litzenberger density, calendar finite-difference). Here $\mathbb{E}[\nu \mid z]$ is the *spot-conditional* average of the unit factor, distinct from the stationary $\mathbb{E}_\pi[\nu] = 1$: near 1, tilted only by the spot–vol correlation. Either way, (17) is the exact discrete Gyöngy projection. The decoupling is then literal: SANOS sets the statics and, through the leverage, the local-vol roll; the kernel carries the SV response (the bulk of the SSR level). In parameter terms the decoupling *is* the split of the fused per-step variance into a market level and a unit-mean kernel shape,

$$\mathrm{Var}\big(\Delta z \mid z, f, s\big) = \underbrace{\sigma^2_{\mathrm{LV}}(z, T_j)\,\Delta_j}_{\text{level (market)}} \;\cdot\; \underbrace{\nu_{f,s}}_{\text{shape (kernel), } \mathbb{E}_\pi[\nu]=1}, \tag{18}$$

whose π-average is exactly $\sigma_{\mathrm{LV}}^2\Delta_j$: the *level* is set by the leverage, and the kernel enters only through the unit-mean ratio $\nu_{f,s} = \bar{V}_{f,s}/\mathbb{E}_\pi[\bar{V}]$. The kernel's own level $\bar{\gamma}$ (which scales every $V_{f,s,l}$ by $e^{\bar{\gamma}}$) nearly divides out of ν and is reset by σ_{LV}, leaving it nearly inert (the $\mathrm{Var}_l(d)$ term breaks exact homogeneity). What survives is the regime-*relative* structure—the $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$ spread of $\bar{V}$ and, through $P_{\mathsf{F}}, P_{\mathsf{S}}$, the leverages $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ and rates $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$—i.e. the SSR (numerically, a unit shift in $\bar{\gamma}$ moves ν by $< 0.2\%$, in ν_{F} by tens of percent). In a *spot-uncorrelated* state $\mathbb{E}[\nu \mid z] \approx 1$: the leverage barely rescales the kernel ($\sigma_{\mathrm{LV}} \approx \sigma_{\mathrm{Dupire}}$), the smile is matched for free, and the SSR is the lone free dynamic target.

Because $\sigma_{\mathrm{LV}}(z)$ is a smooth deterministic statics object, collocating it at the component means affects only the level, not the (translation-invariant, exact) SSR mechanism. For *marginal matching*—each step re-seeded from the target law, regimes at their stationary spread—the residual is discretisation, not bias: the discrete Dupire under-reads the local variance at the narrowest short-T marginals ($\sim$50–85 bp), tightening with a central-difference Dupire, finer grids, and a tail-clamped leverage. Under multi-step *concatenation from a fixed spot* (forward densities, hedging; Section 4.7) the shortcut is *not* free: the negative leverage tilt correlates high-ν regimes with low z, so $\mathbb{E}[\nu \mid z{=}0]$ falls below 1 ($\approx$0.7 on SPX) and $\sigma_{\mathrm{LV}} \approx \sigma_{\mathrm{Dupire}}$ under-reads the accumulated ATM level by $\sim$10% in vol over a multi-month chain. There the exact projection (17) is retained: $\mathbb{E}[\nu \mid z]$ is negligible for reproducing marginals but a genuine path-conditional rescaling for absolute-level outputs (forward implied vol, hedge P&L).

4.7 Recompression: keeping the budget flat under concatenation

Recompression is a *multi-step* device: a *single* kernel application leaves a manageable $n_{\mathsf{L}}n_{\mathsf{F}}n_{\mathsf{S}}$-component mixture and needs none. It is *concatenation*—composing $\mathcal{K}_j \circ \mathcal{K}_{j+1} \circ \cdots$ over several steps for forward densities, exotics, or interpolation—that compounds the count. Each composed step multiplies the per-fibre spot mixture by $n_{\mathsf{L}}n_{\mathsf{F}}n_{\mathsf{S}}$: each target fibre aggregates the $n_{\mathsf{F}}n_{\mathsf{S}}$ source regimes and the n_{L} within-step nodes. Left alone, the count is $(n_{\mathsf{L}}n_{\mathsf{F}}n_{\mathsf{S}})^J$ after J steps. Each composed step therefore ends with a recompression that holds the budget flat: it preserves the *propagated* marginal with fewer components. The match to the SANOS *target* is the separate Gyöngy step (Section 3.3), not part of recompression. The default is a deterministic *moment-preserving merge*: per fibre, sort by mean, split into n_{keep} equal-weight chunks, and match each chunk's first three moments. One may equivalently refit a compact martingale kernel $\widetilde{\mathcal{K}}_j$ reproducing the propagated marginal at the digital level,

$$\min_{\widetilde{\mathcal{K}}_j} \; d_{\mathrm{CDF}}\big(\rho_j\widetilde{\mathcal{K}}_j, \rho_j\mathcal{K}_j\big) \quad \text{s.t.} \quad \widetilde{\mathcal{K}}_j \geq 0, \; \textstyle\sum_l \widetilde{w}_l \, e^{\widetilde{d}_l + \frac{1}{2}\widetilde{\sigma}_l^2} = 1 \text{ (per fibre)}. \tag{19}$$

Because $\widetilde{\mathcal{K}}_j$ is a martingale kernel, $\rho_j\widetilde{\mathcal{K}}_j \succeq_{\mathrm{cx}} \rho_j$ automatically (Strassen). Convex order—hence calendar no-arbitrage—therefore holds by construction, and the convex-order inequalities need not be imposed. The martingale is re-locked by (13) after the merge, which preserves the moments of z but not the forward $\mathbb{E}[e^z]$. Where the merge is too coarse, a constrained weight-LP is available as a refinement: fixed reduced locations, weights fitted to the propagated CDF subject to exact forward and convex order, with the residual reported. Recompression is done every composed step, per-fibre (never merging away the carried regimes), deterministically, with the residual reported.

Recompression touches only the component *budget*; the marginal-match is restored *structurally*, by re-applying the Gyöngy leverage (17) to the recompressed mixture. The leverage is a local, deterministic rescaling read from the recompressed mixture's own Dupire surface against the SANOS target. It therefore absorbs the moment-merge error in the same operation that matches the kernel's intrinsic marginal. The per-step pipeline is *propagate* (dynamics) $\rightarrow$ *recompress* (budget +

re-lock) $\to$ *leverage* (Gyöngy) $\to$ re-lock, with *no projection at any step.* Convex order survives because the match is to the convex-ordered SANOS target, and the leverage edits only the local *scale*, leaving the regime dynamics—the SSR—exact.

4.8 Intermediate maturities: OU-semigroup interpolation

Because the carried factor is OU, the generator is time-homogeneous and the transition operators form a semigroup. Write the latent log-variance as $x_t = g_0(t) + X_t^{\mathsf{F}} + X_t^{\mathsf{S}}$: a time-inhomogeneous forward-variance level g_0 plus a time-homogeneous generator. One then interpolates only the one-dimensional curve g_0, and constructs intermediate kernels by evaluation. At the midpoint the kernel square root is the half-time semigroup element $\mathcal{K}_{[T_1,T_2]} = P_{\Delta T/2}$, so $\mathcal{K} \circ \mathcal{K} = P_{\Delta T} = \mathcal{K}_{[T_1,T_3]}$ exactly, with no optimisation. The carried vol state is what makes this work. The spot-only kernel—marginalising the vol state at the intermediate step—does not compose, which is why a memoryless kernel requires a matching optimisation. Calendar no-arbitrage on the interpolated marginal is automatic when g_0 gives non-decreasing total variance.

5 The fitting step: dynamic targets and the digital metric

The kernel carries the *dynamics*, so it is calibrated to its two dynamic readouts—the SSR term structure and the vol-of-vol. The marginals are not fitted here: the leverage carries them by construction (Section 3.3). What *is* measured against the market—the marginal residual after the leverage, and the SANOS fit in the static layer—is measured in *digitals* (CDFs), not prices. A digital is the strike-derivative of a call, $-\partial_K C = 1 - G(K)$: one derivative above the price, one below the density. The bound that justifies that choice lives in the *spot/strike* variable s (with G the law of S_T):

$$\begin{aligned} \sup_K |C_{\boldsymbol{\theta}} - C_\star| \leq \; & \|C_{\boldsymbol{\theta}} - C_\star\|_{\dot{W}^{1,1}} = \int_0^\infty |G_{\boldsymbol{\theta}}(s) - G_\star(s)|\,\mathrm{d}s \\ = \; & W_1(\rho_{\boldsymbol{\theta}}, \rho_\star) = \int_0^1 |G_{\boldsymbol{\theta}}^{-1}(u) - G_\star^{-1}(u)|\,\mathrm{d}u. \end{aligned} \tag{20}$$

The middle equality is what makes the digital fit a *price* metric. The L^1 norm of $\partial_K(C_{\boldsymbol{\theta}} - C_\star)$ is the homogeneous Sobolev $\dot{W}^{1,1}$ seminorm of the call-price difference—its total variation, and, since $-\partial_K C$ is the digital, the *aggregate mispricing of all digitals.* So measuring a marginal gap $G_\star - G_{\boldsymbol{\theta}}$ digital by digital, as the band-loss (21) does, is a price-space Wasserstein distance—the right metric whether it fits the SANOS marginals (the static layer) or, in the kernel step, bounds their residual after the leverage. The digital is the natural target because it sits one strike-derivative below the noisy density and one above the coarse price.

Remark 1 (Coordinate caveat)**.** *The call payoff $(s-K)^+$ is 1-Lipschitz in spot s, which is why the sup price gap is bounded by W_1 in spot. In log-coordinate $z = \log s$ the integral acquires the Jacobian weight, $\int |G_{\boldsymbol{\theta}} - G_\star| e^z\,\mathrm{d}z$, so an unweighted log-space CDF loss does not by itself control call prices. Accordingly, any log-space, exponential-growth discrepancy used adversarially below is a financial pricing discrepancy / integral probability metric, not literally Wasserstein-1. We reserve "Wasserstein" for the spot-coordinate bound* (20)*.*

Objective. Collect the generator parameters $\boldsymbol{\theta}$. The objective stacks the two dynamic penalties on the SPX digital residual:

$$\begin{aligned}\min_{\boldsymbol{\theta}} \sum_{i\in\mathrm{SPX}} w_i^{\mathrm{spr}} \operatorname{dist}\big(1-G^S_{\boldsymbol{\theta}}(K_i), [\underline{S}_i, \overline{S}_i]\big)^2 \;+\; \beta_{\mathrm{SSR}} \sum_T \big(\mathrm{SSR}_T(\boldsymbol{\theta}) - \mathrm{SSR}^\star_T\big)^2 \;+\; \beta_{\mathrm{vov}}\, \mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta}) \\ +\; \beta_{\mathrm{mono}} \sum_T \Big[\frac{\mathrm{SSR}_{T^+}(\boldsymbol{\theta}) - \mathrm{SSR}_T(\boldsymbol{\theta})}{\mathrm{SSR}_T(\boldsymbol{\theta})}\Big]_+^2 \;+\; \sum_k \varrho_k \big(\theta_k - \theta^{\mathrm{a}}_k\big)^2.\end{aligned} \tag{21}$$

The last two terms are the shape and regularisation controls of the fitting protocol (Section 7.2). Because $[\,\cdot\,]_+$ is one-sided, the monotonicity term is *inert* on a decreasing model SSR term structure and active only where it rises with tenor (β_{mono}=3, applied model-side, never to the target); the ridge is relative and pulls towards the stage anchor $\boldsymbol{\theta}^{\mathrm{a}}$, weighted more weakly on the identified directions than on the nuisance ones. In the reported fits the digital-band weight is set to *zero*—the Gyöngy leverage already fixes the density, and up-weighting the band degrades the SSR fit by an order of magnitude (Section 7.2)—so the kernel is fit to the dynamic targets under these two controls. Here $G^S_{\boldsymbol{\theta}}$ is the CDF of the model's terminal law under the kernel $\boldsymbol{\theta}$—the *Gyöngy-fused* propagated marginal, whose leverage (17) carries the SANOS statics. Writing it as the mixture $\sum_i q^i_{\boldsymbol{\theta}}\, \mathcal{N}(\,\cdot\,; \mu^i_{\boldsymbol{\theta}}, (\sigma^i_{\boldsymbol{\theta}})^2)$ in log-moneyness, with component forwards $F^i_{\boldsymbol{\theta}} = F e^{\mu^i_{\boldsymbol{\theta}} + \frac{1}{2}(\sigma^i_{\boldsymbol{\theta}})^2}$, the band-loss uses the model digital

$$1 - G^S_{\boldsymbol{\theta}}(K_i) = \Pr{}_{\boldsymbol{\theta}}(S > K_i) = \sum_i q^i_{\boldsymbol{\theta}}\, \Phi(d^i_2), \qquad d^i_2 = \frac{\log(F^i_{\boldsymbol{\theta}}/K_i) - \frac{1}{2}(\sigma^i_{\boldsymbol{\theta}})^2}{\sigma^i_{\boldsymbol{\theta}}},$$

the Breeden–Litzenberger survival $-\partial_K C_{\boldsymbol{\theta}}$ of the mixture call (3), fit to the market digital bid–ask band $[\underline{S}_i, \overline{S}_i]$. The w_i^{spr} are inverse-spread weights, and $\beta_{\mathrm{SSR}}, \beta_{\mathrm{vov}}$ the soft-target weights. Here $\mathrm{SSR}_T(\boldsymbol{\theta})$ is the model SSR (22) (in closed form, (23)) and $\mathrm{SSR}^\star_T$ its target, while $\mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta})$ is the vol-of-vol readout residual (30). Both *dynamic* targets—the SSR and the vol-of-vol readout—are developed in Section 6. The minimisation is subject to the admissibility constraints (7). The martingale property is one of them, enforced *exactly* rather than softly. A single-marginal mixture calibration would add a forward-matching cost $\lambda\big(\sum_k \pi_k e^{m_k + q_k/2} - 1\big)^2$ to the loss. Here the per-fibre lock (13) drives the conditional forward error to zero at every state and every $\boldsymbol{\theta}$, so the analogous term is identically zero and is dropped. The digital band-loss is likewise not a fit of the statics but the *residual* of the leverage's structural match (17), holding the small collocation and discrete-Dupire discrepancy inside bid–ask. What $\boldsymbol{\theta}$ genuinely controls is the pair the translation-invariant kernel is dense in (Appendix C): the SSR and the vol-of-vol.

Two regimes for the dynamic targets. Where forward-start and VIX digitals are *liquid*, fit those prices directly through the band-loss and let the SSR *emerge*, validating it against the empirical value as a diagnostic—the principled, risk-neutral route. Even then the emergent SSR is not guaranteed: the Quintic OU model fits the full SPX *and* VIX smiles jointly and still finds its SSR significantly below the market's, restoring it with exactly such a band penalty [28]. Where the dynamic instruments are thin (most underlyings), the β_{SSR} penalty instead acts as an interpretable *steer*. Its target is a realized-SSR estimate or a desk view: a real-world ($\mathbb{P}$) quantity, not the risk-neutral SSR, so it neglects the volatility risk premium. The penalty form is robust either way; an adversarial form (a min–max over an exponential-growth critic for the worst-mispriced digital) is an optional alternative when forward-start/VIX digitals are rich.

Algorithm 1 orchestrates the full daily calibration, with the SANOS LP (Algorithm 2) as its static first step. The kernel fit is the one *non-convex* part—in contrast to the convex SANOS

LP—and the structural marginal-matching (Section 3.3: the leverage (17) carries the statics, then moment-preserving recompression re-locks the martingale) guarantees the returned model is exactly arbitrage-free wherever the optimiser lands: its propagated marginals are martingale-coupled and convex-ordered by construction, so the surface admits no static arbitrage regardless of where the fit lands. What the optimiser controls is the *accuracy* of the match to the SANOS targets—the residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$ reported in Section 7.2 and held within the quote bands—not the arbitrage-freeness.

Algorithm 1 SANOS-Evolve calibration (one pricing date).

Require: per-maturity option chains (calls and puts) $\{(T_j, K, \text{bid/ask})\}_{j=1}^{M}$ and a discount curve (the static layer); *and*—since the chains pin the marginals but not the dynamics—a dynamic anchor for the SSR/vol-of-vol: forward-starts or a vol-index (VIX for SPX) where liquid, else a realised estimate from the underlying's history. The forward curve (put–call parity) and forward-variance level (variance swap) are derived from the chains

Ensure: marginals $\{\mu_j\}$ and generator $\boldsymbol{\theta} = (\bar{\gamma}, \nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}, \lambda_{\text{skew}}, \lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}, \kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$ defining admissible kernels $\{\mathcal{K}_j\}$; the desk outputs of Table 1

1: **Static layer.** For each T_j, fit μ_j by the SANOS LP (Algorithm 2); verify increasing convex order $\mu_j \preceq_{\text{cx}} \mu_{j+1}$ (the precondition of Theorem 1); repair the marginals if violated.

2: **Readout.** Extract the vol-of-vol level $(v_\infty, \xi_{\text{VIX}})$ from the forward-variance readout—VIX futures and call-spreads for SPX, else the strip's variance-swap level with smile curvature and realised vol-of-vol (Table 2); the two timescales $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$ are *both* fit, their fast/slow split resolved by the VIX vol-of-vol *term structure* (the damping $D(\tau)$ at two horizons) on top of the SSR slope (Appendix D).

3: **Regime scaffold.** Fix Gauss–Hermite nodes $\{\zeta_f^{\mathsf{F}}\}, \{\zeta_s^{\mathsf{S}}\}, \{\zeta_l^{\mathsf{L}}\}$ and base log-weights η; initialise $\boldsymbol{\theta}$ from a global vol-quantile split and the readout.

4: **for** $j = 1, \ldots, M-1$ (warm-started across maturities) **do**

5:     **repeat**

6:         **Kernel fit.** Minimise the objective (21) over $\boldsymbol{\theta}$: digital band-loss to μ_{j+1}, plus β_{SSR} (SSR term structure) and β_{vov} (vol-of-vol readout), subject to admissibility (7) (positivity, normalisation, exact martingale lock). ▷ non-convex

7:         **Structural match + recompress.** The leverage (17) lands the forward map on μ_{j+1} by construction; moment-preserving recompression (Sec. 4.7) holds the component budget and re-locks the martingale. Record the residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$.

8:     **until** marginal, SSR, and vol-of-vol residuals are within their bands

9: **end for**

10: **return** $\{\mu_j\}$, $\boldsymbol{\theta}$, $\{\mathcal{K}_j\}$; price vanillas/exotics, Greeks, and SSR-deltas in closed form (Table 1).

What lands on the desk. The kernel is an internal engine behind a standard vol-surface API; the desk receives the deliverables of Table 1, all closed-form and all functionals of the *same* calibrated object the algorithm returns. Continuously monitored barriers are the one caveat—they need sub-stepping or a bridge correction—while the other exotics are direct kernel functionals.

6 Smile dynamics: the SSR and the forward-variance readout

This section develops the two *dynamic* targets of the calibration objective (21): the skew-stickiness ratio $\text{SSR}_T(\boldsymbol{\theta})$ (the regime comovement, in closed form below) and the forward-variance / vol-of-vol

Table 1: What lands on the desk: every output is a functional of one calibrated object (SANOS marginals + the transition kernel).

Deliverable	Source
SPX vanilla surface + Greeks	spot marginals (drop-in)
Forward-vol surface $\widehat{\Sigma}^{\mathrm{fwd}}(z,k)$	transition kernel
VIX-identified vol-of-vol	kernel (VIX-informed, Sec. 6)
Exotics: forward-starts, cliquets, autocallables, barriers	propagate through the kernel
SSR-consistent Δ, vanna	kernel dynamics (Sec. 6)

readout $\mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta})$. Each is a closed-form function of the generator $\boldsymbol{\theta}$. The desk-facing ATM features and the SSR-consistent hedge follow from the same machinery.

6.1 The skew-stickiness ratio

With $k = \log(K/F)$, ATM skew $\mathcal{S}_T = \partial_k \widehat{\sigma}_T(k)|_{k=0}$, the SSR of Bergomi [22] is

$$\mathrm{SSR}_T = \frac{1}{\mathcal{S}_T}\frac{\mathrm{d}\widehat{\sigma}_{\mathrm{ATM},T}}{\mathrm{d}\log S}. \tag{22}$$

In this convention SSR = 1 is *sticky-strike* (fixed-strike vols pinned; the ATM vol slides along the skew) and SSR = 0 is *sticky-delta* (the smile rides with spot; ATM vol unchanged). The empirical SPX term structure of Section 1 sits at or beyond sticky-strike, never near sticky-delta. We fix this convention throughout.

6.2 The SSR is a closed-form realised covariance

A single marginal pins $\mathrm{Law}(S_T)$ but not how the smile moves; that information lives in the kernel. Because the kernel is translation-invariant in z (Section 3), the conditional smile $\sigma(T; f, s)$ depends only on the regime pair. The empirical SSR—the regression of ATM-vol changes on returns—is therefore *not* a spot-derivative but a finite closed-form sum over the regime transitions. With $\pi_{f,s}$ the stationary law of the carried chain, r the one-step return with node values $d_{f,s,l}$, and $\mathrm{skew}(T) = \sum_{f,s} \pi_{f,s}\, \mathcal{S}_T(f,s)$ ($\mathcal{S}_T(f,s) = \partial_k \sigma(T; f, s)|_{k=0}$ the regime-conditional ATM skew),

$$\begin{aligned}
\mathrm{SSR}(T) &= \frac{\mathrm{Cov}\left(\Delta\sigma_{\mathrm{ATM}}(T),\, r\right)}{\mathrm{Var}(r)\,\mathrm{skew}(T)},\\
\mathrm{Cov} &= \sum_{f,s}\pi_{f,s}\sum_{l} w_l\, d_{f,s,l}\sum_{f',s'} P_{\mathsf{F}}(f'|l,f)P_{\mathsf{S}}(s'|l,s)\left[\sigma(T;f',s') - \sigma(T;f,s)\right],\\
\mathrm{Var}(r) &= \sum_{f,s}\pi_{f,s}\Big[\sum_{l} w_l(d_{f,s,l}^2 + V_{f,s,l}) - \big(\sum_{l} w_l d_{f,s,l}\big)^2\Big],
\end{aligned} \tag{23}$$

with $\sigma(T; f, s)$ the ATM vol of the (T/Δ_j)-step propagation started in regime (f, s), and $\mathrm{Var}(r)$ the π-averaged conditional return variance. The between-regime mean term $\mathrm{Var}_\pi(\sum_l w_l d_{f,s,l}) \sim 0.2\%$ is higher-order and dropped. The covariance *is* the leverage: a down-return node l tilts $P_{\mathsf{F}}, P_{\mathsf{S}}$ toward higher vol (via $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$), so $\Delta\sigma$ correlates with r. It is a finite sum over (l, f, s, f', s'): no spot-bump, no Monte Carlo, no measure-transport approximation. It is also Monte-Carlo-confirmed—the sum sits at the centre of a direct one-step regression, $|\mathrm{diff}| \approx 0.005$ at 1m/3m/6m/1y (Section 7.1). Once the (17) leverage is in, the realised SSR picks up a second, additive piece: a *local-vol roll* $\partial_z\hat{\sigma}$, the spot-shift derivative $\mathrm{d}\hat{\sigma}_{\mathrm{ATM}}/\,\mathrm{d}\log S$ with $z=\log(S/F)$. It is set by the forward-variance curve, which is statics owned by SANOS. This sits on top of the stochastic-vol response, which carries the bulk of the level. Both pieces are closed form, and the fused readout is likewise Monte-Carlo-validated.

6.3 The SSR dials

The innovation leverages $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ carry the comovement (Section 4.4): $\lambda_{\mathsf{F}}=\lambda_{\mathsf{S}}=0 \Rightarrow \mathrm{SSR}=0$, and raising them lifts it through the sticky-strike value $\mathrm{SSR}=1$ toward the short-maturity diffusive limit near 2. The empirical SPX range $\mathrm{SSR}\in[1,2]$ is therefore a leverage-magnitude statement. Acting through the two timescales with $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$, they shape the SSR *term structure*—level *and* slope. The fast κ_{F} gives the high, decaying short end; the slow κ_{S} the sustained long-end floor that one factor cannot. That is what Section 5 fits to SPX. (The SSR is regime- and vol-of-vol-dependent rather than fixed by the variance-curve slope. A rising forward-variance curve lifts its *level* relative to a flat one [25]—a leading-order, qualitative effect, not a slope law.) Where forward-start prices are liquid, the same (λ,κ) are pinned by fitting those prices directly. The kernel prices forward-starting calls in closed form, $C^{\mathrm{FS}}_{T_1,T_2}(k)=D_{T_2}\int\rho_{T_1}(\,\mathrm{d}x)\int\mathcal{K}_{T_1,T_2}(x,\,\mathrm{d}y)(e^{y-x}-k)^+$, Its strike-derivative is the forward-return survival function, the clean traded proxy for the SSR, so the SSR *emerges* as a diagnostic. How much the SSR can move at *fixed* smile is bounded: this is the decoupling band of Section 7.1, the soft direction of Appendix D made quantitative.

6.4 Closed-form ATM features and the SSR

Because the conditional forward-return law is a Gaussian (log-normal) mixture, the ATM implied-vol Taylor coefficients—level $\hat{\sigma}(0)$, skew $\hat{\sigma}'(0)$, curvature $\hat{\sigma}''(0)$—are closed form *with no Black–Scholes inversion*, by two routes.

Exact: implicit differentiation. The implied vol solves $C_{\mathrm{mix}}(m)=C_{\mathrm{BS}}(m,\hat{\sigma}(m))$ in log-moneyness $m=\log(F/K)$ (so $\partial_m=-\partial_k$ for the §6 skew variable k; sign in App. E). The level is a single probit of the closed-form ATM mixture price. The higher coefficients follow by implicitly differentiating that identity (derived in full in Appendix E), working in the *total* volatility $u=\hat{\sigma}\sqrt{T_j}$ (the variable the Black greeks below are taken in),

$$u'(0)=\frac{\partial_m C_{\mathrm{mix}}-\partial_m C_{\mathrm{BS}}}{\mathcal{V}},\qquad u''(0)=\frac{\partial_m^2 C_{\mathrm{mix}}-\partial_m^2 C_{\mathrm{BS}}-2\,\mathcal{V}_x\,u'(0)-\mathcal{V}_u\,u'(0)^2}{\mathcal{V}},\qquad \hat{\sigma}^{(k)}(0)=\frac{u^{(k)}(0)}{\sqrt{T_j}},\tag{24}$$

with the *total-vol* Black–Scholes greeks vega $\mathcal{V}$, vanna $\mathcal{V}_x$, and volga $\mathcal{V}_u$ (u-derivatives of C_{BS}) and $\partial_m^k C_{\mathrm{mix}}$ the closed-form strike-derivatives of the mixture price. The last identity annualises each coefficient. These are exact, numerically confirmed against the bisection-inverted IV in App. E. The SSR $\mathrm{SSR}=\partial_z\hat{\sigma}(0)/\hat{\sigma}'(0)$ matches the closed-form realised covariance (23) to four decimals. We take (24) as the calibration targets.

Leading order: cumulants. A coarser but interpretable route reads the coefficients off the regime-conditional mixture cumulants $\kappa_n=\partial_t^n L(0)$ of the cumulant-generating function

$$L(t)=\log\sum_l w_l\,e^{t\,d_{f,s,l}+\frac{1}{2}t^2 V_{f,s,l}},\tag{25}$$

via the Gram–Charlier map $u_0=\sqrt{\kappa_2}$, $u_1=\kappa_3/6\kappa_2^{3/2}$, $u_2=\kappa_4/12\kappa_2^{5/2}$ [11, 12]. By the law of total cumulance $\kappa_3=\mu_3(d)+3\,\mathrm{Cov}(d,V)$, the skew's leverage term $\mathrm{Cov}(d,V)$ is explicit and the

Appendix D Jacobian is analytic; the SSR itself is the realised covariance (23), not a spot-derivative of these regime-conditional cumulants. The cost is accuracy—this proxy sits 8–9% below the exact level (skew 16–20%)—so the cumulants serve interpretation and identifiability, not the fit.

Roughness consistency, and the SSR as a calibration map. The inverted $\boldsymbol{\theta}$ should respect the static-to-dynamic relation SSR $\to H + \frac{3}{2}$ when the skew decays as $\tau^{H-1/2}$ [33], with H the Hurst exponent of the volatility process [25, 26]. Read backwards this is a calibration rule: the SSR is the trusted observable, so an observed short-T value fixes the model's skew-dynamics exponent ($H = \mathrm{SSR}^\star - \frac{3}{2}$) rather than leaving it free. The diffusive two-OU kernel sits at $H = \frac{1}{2}$ (SSR $\to 2$); a market SSR ≈ 1.5 reads as $H \approx 0$—the roughness a multi-factor extension must hit, a derived target, not a knob.

6.5 The SSR-consistent hedge ratio

The desk-facing payoff is the hedge. The true delta of a strike-K option carries a smile-response term, and for a *fixed* strike the sensitivity is

$$\Delta_{\mathrm{SSR}} = \Delta_{\mathrm{BS}} + \mathrm{Vega} \cdot \partial_S \widehat{\sigma}(K), \qquad \partial_S \widehat{\sigma}(K) = \frac{\mathcal{S}_T\,(\mathrm{SSR}_T - 1)}{S}, \tag{26}$$

where $\mathcal{S}_T = \partial_k \widehat{\sigma}$ is the ATM skew. The factor $(\mathrm{SSR}_T - 1)$ combines the SSR's at-the-money response, $\partial_S \widehat{\sigma}_{\mathrm{ATM}} = \mathrm{SSR}_T\,\mathcal{S}_T/S$, with the fixed-strike smile-roll $-\mathcal{S}_T/S$. Hence sticky-strike (SSR $= 1$) gives $\partial_S \widehat{\sigma}(K) = 0$ and recovers the Black delta, while sticky-delta (SSR $= 0$) gives $-\mathcal{S}_T/S$. A Black/sticky-strike hedge omits the $(\mathrm{SSR}_T - 1)$ term. It bleeds systematic P&L whenever the realised SSR departs from 1—materially so for SPX at short tenors—while the calibrated kernel supplies the adjustment its calibrated SSR implies. We name this delta *SSR-consistent* rather than "correct": (26) is the analytically correct hedge *for a smile rolling at the calibrated SSR*. Whether it lowers realised hedging variance against the Black and sticky-strike deltas is a separate, empirical claim. The historical replay of Section 7 finds it beats Black by 30–50% at the money (the expected minimum-variance-delta effect) and the industry minimum-variance/Hull–White delta more modestly, with the broader multi-moneyness study outstanding.

The hedging error. What (26) buys is the residual it removes. For a Black-delta hedge ($\Delta = \Delta_{\mathrm{BS}}$) of a strike-$K$ option, the step P&L decomposes as

$$\mathrm{d}V - \Delta_{\mathrm{BS}}\,\mathrm{d}S = \underbrace{\mathrm{Vega} \cdot \frac{\mathcal{S}_T(\mathrm{SSR}_T - 1)}{S}\,\mathrm{d}S}_{\text{smile-roll (first order)}} + \underbrace{\tfrac{1}{2}\Gamma(\,\mathrm{d}S)^2 - \Theta\,\mathrm{d}t}_{\text{gamma–theta}} + \cdots, \tag{27}$$

the smile-roll term being the value change from the smile moving with spot ($\mathrm{d}\widehat{\sigma}(K) = \partial_S \widehat{\sigma}(K)\,\mathrm{d}S$). It is a *first-order* exposure, systematic whenever $\mathrm{SSR}_T \neq 1$, that inflates the hedging-error variance over many rebalances. The SSR-consistent delta (26) folds it into the spot hedge, leaving the usual gamma–theta. The companion vanna—the spot–vol cross-greek, equally closed-form—hedges the second-order cross: under the SSR the realised spot–vol comovement is $\mathrm{SSR}_T\,\mathcal{S}_T(\,\mathrm{d}S)^2/S$, an exposure the calibrated SSR likewise pins.

6.6 The forward-variance readout (VIX as the SPX instance)

The vol-of-vol is *identified* from a readout of the underlying's forward-variance distribution, and enters the fit only as a soft target. Its *level* is the variance-swap rate: a model-free integral of the underlying's own option strip, hence available for *any* European-option underlying with no separate product. Its *dispersion* (vol-of-vol and skew) is what a volatility-index options market prices directly. For SPX that market is VIX, the sharpest instance. For SPX, VIX_T^2 is 30-day forward variance, $Y_T := \mathrm{VIX}_T^2 = \frac{1}{\tau}\mathbb{E}_T\big[\int_T^{T+\tau} v_u\,\mathrm{d}u\big]$ with $\tau \approx 30/365$, a functional of the latent vol state. A factor with rate κ is damped over the readout horizon by $D(\kappa) = (1 - e^{-\kappa\tau})/(\kappa\tau)$. The fast factor is strongly damped and the slow factor barely, so the readout is slow-dominated while short-dated smile curvature is fast-dominated. With $V^{\mathsf{F}}, V^{\mathsf{S}}$ the fast/slow factor contributions to the forward variance—affine readout $Y_T = v_\infty + D_{\mathsf{F}}V^{\mathsf{F}} + D_{\mathsf{S}}V^{\mathsf{S}}$, $D_{\mathsf{F}} = D(\kappa_{\mathsf{F}})$, $D_{\mathsf{S}} = D(\kappa_{\mathsf{S}})$—this separates the timescales:

$$\mathrm{Var}(Y_T) = D_{\mathsf{F}}^2\,\mathrm{Var}(V^{\mathsf{F}}) + D_{\mathsf{S}}^2\,\mathrm{Var}(V^{\mathsf{S}}) + 2D_{\mathsf{F}}D_{\mathsf{S}}\,\mathrm{Cov}(V^{\mathsf{F}}, V^{\mathsf{S}}). \tag{28}$$

Remark 2 (Identification caveat)**.** *The clean "two observables, two unknowns" reading—smile curvature and readout width recover* $\mathrm{Var}(V^{\mathsf{F}}), \mathrm{Var}(V^{\mathsf{S}})$*—holds only once the factor correlation* χ_{fs} *is fixed. Our default* $\chi_{fs} = 1$ *pins the covariance term in* (28) *to* $\sqrt{\mathrm{Var}(V^{\mathsf{F}})\,\mathrm{Var}(V^{\mathsf{S}})}$*, keeping two unknowns; with* χ_{fs} *free there is a third.*

In the discrete kernel the forward variance is regime-determined—$\mathrm{VIX}_T = \sqrt{\bar{v}_k}$ on regime k with probability p_k—so the variance-index law is atomic and its calls are regime sums,

$$C^{\mathrm{VIX}}(K) = e^{-rT}\sum_k p_k\big(\sqrt{\bar{v}_k} - K\big)^+, \tag{29}$$

with $\bar{v}_k$ the forward variance carried by regime k: no PDE, no Monte Carlo.

Remark 3 (Atomic vs. dense regimes, and the VIX smile)**.** *The VIX law in* (29) *is atomic because* $\mathrm{VIX}_T = \sqrt{\bar{v}_k}$ *is a deterministic function of the discrete regime; hence the VIX call is piecewise-linear in* K *and the VIX digital is a step function. This is harmless under our scope, where VIX only* identifies *the regime dispersion. The atomic sum is a quadrature for that dispersion, and its accuracy improves with the regime count—a discretisation knob, not a parameter: the generator stays* $\sim$6-*dimensional. If a smooth VIX smile is required, give each regime a within-regime lognormal spread, whereupon* (29) *becomes a mixture of Black calls—at the cost of one vol-of-VIX parameter per regime.*

The readout enters the objective (21) through a soft target on the kernel's implied vol-of-vol, computed from the closed-form regime sums (29):

$$\mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta}) = \big(v_\infty(\boldsymbol{\theta}) - v_\infty^\star\big)^2 + \sum_n \big(\xi_{\mathrm{model}}(\tau_n;\boldsymbol{\theta}) - \xi_{\mathrm{VIX}}(\tau_n)\big)^2, \tag{30}$$

the squared mismatch of the model's forward-variance *level* v_∞ and *vol-of-vol* ξ to the readout, summed over the horizons $\{\tau_n\}$. Their term structure pins κ_{S} through the damping $D(\tau)$, and a vol-of-vol-skew term is added where a vol-index smile is liquid. It is *not* a hard marginal. We impose only *unconditional* forward-variance consistency, $\sum_k p_k y_k = \mathrm{FV}$, which is necessary but not sufficient for full joint consistency. The conditional identity $y \approx \mathbb{E}[L_\tau \mid S, Y]$ that the joint problem [31] requires is exactly what reading rather than jointly fitting forgoes.

6.7 Extracting the readout, and what stands in for VIX

The targets ξ_{VIX} and κ_{S} are read by a small, robust estimator rather than a full smile fit, since the readout is a soft target. For SPX the instruments are VIX:

1. **Instruments.** VIX *futures* for the forward-variance level, and VIX *call-spreads*—not pure digitals, as the model variance-index law is atomic (Remark 3)—for the smile width and skew, across the listed maturities $\{\tau_n\}$.

2. **Filtering.** Retain quotes inside liquid bid–ask bands and weight each by inverse spread.

3. **Targets.** Futures $\to$ level / long-run variance v_∞; smile *width* $\to$ vol-of-vol ξ_{VIX}; smile *skew* $\to$ vol-of-vol skew; *term structure* of the width across $\{\tau_n\}$ $\to$ the slow rate κ_{S} via the damping $D(\tau)$.

4. **Map to the generator.** Fix the two timescales $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$ from the term structure of $\xi_{\mathrm{VIX}}(\tau_n)$ through $D(\tau)$ at two horizons; split $\nu_{\mathsf{S}}, \nu_{\mathsf{F}}$ from ξ_{VIX} and the short-dated smile curvature via (28), conditional on χ_{fs}. All enter $\mathcal{R}_{\mathrm{vov}}$ as soft targets.

For an underlying *without* a liquid volatility-index options market the same recipe runs against whatever readout exists, at graded identification strength (Table 2): the target is never empty, merely sharper for SPX. Off-SPX the readout is *weakly* identified rather than absent: the own-strip skew and curvature give ρ and ξ separately but *fuzzily* (the soft pair of Appendix D), still risk-neutral but noisy. A realised estimate optionally firms it up, at the cost of mixing in a real-world ($\mathbb{P}$) quantity. VIX simply makes the same $\mathbb{Q}$ separation sharp—the honest cost of generality. The vol-of-vol *skew* (the upward VIX-smile slope) is third-order for the SSR and near-ATM dynamics and has no off-SPX market source, so the parsimonious kernel omits it. It is fitted only for VIX-pricing or deep-wing work, where SPX's liquid VIX makes it available.

Underlying	Forward-variance readout	Strength
SPX	VIX futures + options	sharpest (market-pinned)
SX5E	VSTOXX futures + options	strong
Crypto (BTC/ETH)	DVOL index + smile + realised	good
NDX, NKY, KOSPI, HSI	vol-index level + smile curvature + realised	workable
FX (EURUSD, ...)	risk-reversal/butterfly + forward-vol	workable
European single names	smile curvature + realised + event structure	weakest
any European underlying	own-strip level + smile curvature + realised	universal fallback

Table 2: The forward-variance readout that identifies the kernel's vol-of-vol, by underlying. The *level* is always recoverable from the option strip; only the *dispersion* (vol-of-vol) varies in identification strength. VIX is the sharpest available instance, not a precondition.

The CBOE VIX is the log-contract, equal to forward variance only without jumps; a diffusive Gaussian-mixture satisfies this, while a jump component needs the standard convexity correction.

7 Empirical study: ground-truth validation and real-data calibration

The study has two halves. On synthetic ground truth—where the generator is known—we verify what the market cannot: that the closed forms are exact, that the statics/dynamics decoupling is a genuine bounded degree of freedom (Section 7.1). The real-data study is the application: real SANOS marginals, the SSR+vov fit across regimes, the off-SPX (NDX) generalisation, and the SSR-consistent hedging replay (Section 7.2).

7.1 Ground-truth validation on synthetic data

All checks run at a fixed reference generator $\boldsymbol{\theta}_0$ (nine knobs, calibrated once to canonical SPX term-structure targets; values in Figure 2), on a synthetic arbitrage-free chain where every quantity is computable both in closed form and by simulation.

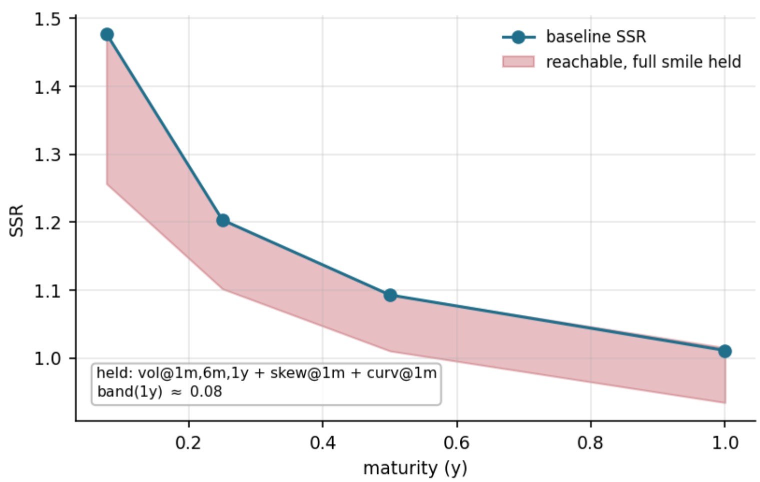

Figure 2: The statics/dynamics decoupling band at the reference generator (closed form, $n_k = 20$). Moving along the steepest *static-preserving* direction—the null-space of the static Jacobian holding *five* smile observables (ATM vol at 1m/6m/1y; skew and curvature at 1m), Newton-held to $\lesssim$ 1 bp in the vols and 10^{-3} in skew—the SSR is freely movable only inside the shaded band: the long end spans $\approx$ 0.93–1.02 (width 0.08) at *fixed* statics. $\boldsymbol{\theta}_0 = (\bar{\gamma}\,-5.30, \nu_{\mathsf{F}}\,0.43, \nu_{\mathsf{S}}\,0.50, \nu_{\mathsf{L}}\,0.14, \lambda_{\text{skew}}\,-1.48, \lambda_{\mathsf{F}}\,0.98, \lambda_{\mathsf{S}}\,1.65, \kappa_{\mathsf{F}}\,1.00, \kappa_{\mathsf{S}}\,2.34)$.

Decoupling and identifiability. The dynamics are a genuine but *narrow* degree of freedom over the statics. Along the steepest static-preserving direction (the null-space of the static Jacobian holding the five smile observables of Figure 2), the SSR(1y) is freely movable only inside a width-**0.08** band $[0.93, 1.02]$ at *fixed* statics. Relaxing the hold to the three-observable smile (ATM vol at 1m/1y, 1m skew) roughly doubles the reach ($\approx [0.89, 1.07]$), which contains the market long-end SSR ($\approx$1.05). The freedom is non-zero (genuine SV, off the local-vol floor) and bounded—wider moves recruit the statics, which the full calibration supplies. The elasticity-scaled Jacobian has effective rank $\approx$7; the closed-form VIX vol-of-vol pins the *timescale* half of the soft direction (a $12\times$ regularisation), leaving the slow leverage λ_{S} for forward-start skew (Appendix D).

Exactness (Monte-Carlo)—the anchor. The closed-form realised SSR sits within Monte-Carlo noise of a direct simulation ($|\text{diff}| \approx 0.005$ at 1m–1y; the fused SLV SSR ≤ 0.007). The check has teeth: an earlier source-averaged form was off by 0.105 on a skewed target, which the Monte Carlo caught. The dynamic readouts are exact closed forms, not approximations, and this is what the real-data residuals are read against. With the machinery proven exact here, the regime-dependence and the small vol-of-vol residual on the lowest-vol chains (Section 7.2) are properties of the data and the model's scope, not numerical error—provided the readouts are taken at converged Gauss–Hermite resolution.

Scope at the short end. The kernel is diffusive ($H=\frac{1}{2}$), so the model SSR rises toward the short-end limit 2; matching SPX's rough sub-month level (≈1.5) needs a rough multi-factor extension (SSR $\to H+\frac{3}{2}$ [26]). At the other extreme, in the lowest-vol regimes (SPX 2017/21/24, NDX 2024) the *realised* 1-week SSR runs *above* the bound (≈2.1–2.5); we report these rather than capping at 2, and the discrete kernel reproduces them by crossing the diffusive ceiling—honest at the data's short end, though those points lie beyond the strict $H=\frac{1}{2}$ scope.

7.2 Real-data study: results

SPX calibration. On real ORATS SPX end-of-day chains the pipeline runs end-to-end. *Statics and propagation:* SANOS marginals fit across ∼20 maturities to ∼2.5 y (convex order holds), and one admissible kernel propagates between adjacent maturities under the exact martingale lock (13). The marginal-propagation residual $d(\mu_j\mathcal{K}_j, \mu_{j+1})$, after the structural Gyöngy match and re-seeded from each real μ_j, is a median ≈ 3 call-price bp and ≈ 26 ATM-vol bp on the liquid surface (≤ 6 m), within bid–ask. It rises to $\lesssim$ 35 call-bp only at the sparse long end, where adjacent expiries are 26–52 steps apart and the per-step leverage-collocation error (§4.6) accumulates—a first-order approximation rather than a convergence dial: the residual is flat in the recompression budget, since the Gyöngy match is re-imposed at every step. *Forward-start:* the forward-start return density is exact (mass and forward equal 1 to machine precision) and its forward skew rolls down with tenor ($\approx -0.5 \to -0.14$), flatter than the spot skew—the smile motion of Section 6 (Figure 3). *Dynamics:* the SSR+vov fit reproduces both the realised skew-stickiness term structure (23) and the VIX vol-of-vol, at ≤3% RMS on both blocks for calm and moderate dates (Table 3, Figures 4–5).[1]

Calibration robustness and the decoupling, empirically. Two features of the fit are worth stating plainly. First, the joint objective is *loosely identified*: the dynamic readouts pin the kernel only up to a multi-basin freedom, and that freedom is not benign—at *identical* readout resolution a cold multi-start and a warm continuation from a neighbouring fit differ by ∼1 point of SSR RMS on SPX 2019 (2.6% against 1.6%), comparable to the fit quality being reported. The reported fits therefore use a diverse multi-start (a few near-anchor seeds under-optimise) and re-seed from a neighbouring good basin wherever the cold start is visibly poorer, under the acceptance rule of the fitting protocol below; the years where this was needed are flagged in the *fit* column of Table 4. They are the best of a documented search, not certified global optima. What is loosely identified is the *parameter vector*, not the priced dynamics: any two fits the protocol accepts reproduce the same SSR and skew term structures to within the reported RMS, the residual basin freedom lying in the $\boldsymbol{\theta}$-directions those readouts leave soft (Appendix D)—the decoupling seen from the calibration side. The object carried downstream is the readout, not $\boldsymbol{\theta}$; the implied SSR that drives the hedge is stable out of sample (sd 0.07, Figure 11). Second, the statics/dynamics decoupling of Section 3.3 holds *regime-independently.* With the digital band given zero weight, so the kernel is fit to the dynamics alone, the from-spot leveraged marginal (the multi-step accumulation from z=0, looser than the re-seeded per-step residual above) still matches the SANOS marginals to ≈5% survival RMS in calm, high-vol and grind alike—the leverage carries the statics whatever the dynamics do. Up-weighting the band into the kernel's objective, by contrast, degrades the SSR fit by more than

[1] The SSR and vol-of-vol targets are estimated over all trading days of the calibration year. Removing days adjacent to scheduled macro releases shifts the estimated SSR term structure by a median 9% (max 15% across 2015–2024), almost entirely through β—the mean skew moves by under 1.5%. The target therefore embeds scheduled-event risk by construction, which is the intended scope: the kernel is fit to the dynamics the market actually prices, not to a release-free counterfactual.

an order of magnitude (the kernel spends its freedom on the density the leverage already fixes), which is why the band enters the loss only as the leverage's structural-match residual (§5).

Diagnostic	Result (real SPX)	Target
Marginal residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$, post-Gyöngy	3 call-bp / 26 IV-bp ($\leq$6 m)	within bid–ask
SSR_T fit (calm/moderate)	0.7–2.7% RMS	realised $1.9 \to 1.6$
VIX vol-of-vol ξ (soft target)	1–6.5% RMS (largest in low-vol yrs)	VIX-implied
Forward-smile	valid density; skew roll-down	—

Table 3: Real-SPX diagnostics; the multi-start SSR+vov fit over 2012–2024 (autodiff Jacobian, ridge regularisation, cached target).

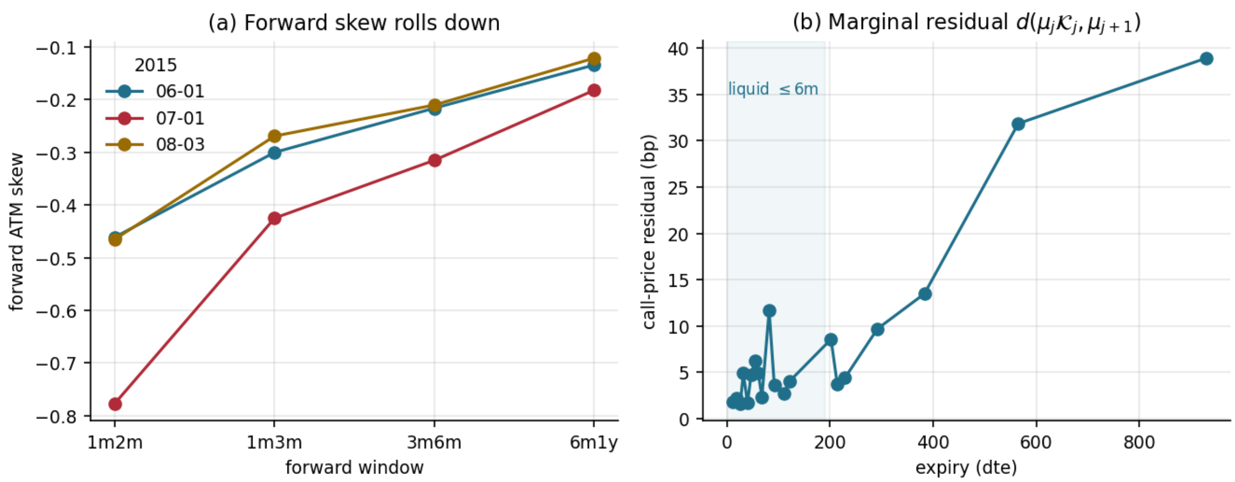

Figure 3: Real-SPX statics and forward-start validation. **(a)** The forward-start ATM skew rolls down (flattens) with tenor across dates—the smile motion of Section 6. **(b)** The marginal-propagation residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$ (2015-06-01), measured *after* the structural Gyöngy match (17), which imposes $\mu_j \mathcal{K}_j = \mu_{j+1}$ by construction—so this is the discrete construction's residual, not a fitted miss: a few call-bp on the liquid surface, growing only at the sparse long end where adjacent expiries are many kernel steps apart. (The realised SSR and VIX vol-of-vol term structures the fit reproduces are shown for all nine years in Figures 4–5.)

Realised-SSR data cleaning. The realised SSR is a $\mathbb{P}$ regression on ORATS end-of-day chains, and three prints must be filtered before it is trustworthy. (i) The day spot is the *median* `stockPrice` across the day's smiles—ORATS records it per option row and it is unreliable (it varies across strikes and is corrupt up to +20% in 2022–23), so any single row injects spurious ±20% one-day returns. (ii) An index smile is always negative-skew, so a day with a positive *fitted* ATM skew is a corrupt short-dated quadratic (thin options) and is dropped, per maturity. (iii) Thin weeklies—pronounced on NDX—occasionally print an absurd short-dated ATM vol (20–45%, up to 2.5× the one-month, on *calm* days), which survives (ii) because its skew is negative; we drop a short maturity whose ATM vol exceeds 1.5× the one-month (a real crash lifts the whole curve, keeping the 1wk/1m ratio below 1.5, whereas the artifact is an isolated short-end spike). Rules (ii)–(iii) are *inert on* *clean data*: on liquid SPX they remove essentially nothing (floors and fits unchanged, COVID crashes retained), while on NDX 2012/2016/2017 rule (iii) removes 5/24/35 corrupt days, restoring the one-week SSR from 0.75–1.02 to ≈1.4–1.6 and its noise floor from 20–41% to 8–15%. The realised *variance-of-variance* needs an analogous clean. NDX has no vol index and only sparse monthly/quarterly expiries past the front, so a fixed-tenor level series built from the *nearest* expiry inherits a spurious jump each time that expiry rolls—the raw 45/60-day NDX var-of-var blows up to 4–10× its neighbours (e.g. 2021: 45 d = 10.8, 60 d = 3.6, against ≈0.6–0.9 either side). We instead build each tenor by *constant-maturity* interpolation—interpolating total variance $\sigma^2 T$ linearly in maturity to the exact target, as VIX does for its 30-day—and drop residual level-jump outliers (> 6 robust MADs); the 60-day gap-centre, never a listed NDX expiry, is excluded. This gives a smooth five-tenor readout (30/45/90/120/180 d) and is again inert on liquid SPX (which has a full VIX-implied term structure and needs none of this).

Fitting protocol. The SSR target is an ordinary least-squares β (daily ATM-vol changes on returns) at the five tenors 1wk–3m over a calendar-year window, after the cleaning above; NDX 2017 is the sole exception, where a large anti-leverage one-week print survives the filters and we take a Huber-robust β instead. To the relative SSR and vol-of-vol residuals the objective adds a relative ridge toward the anchor and a one-sided, *model*-side monotonicity penalty $\big[(s_{i+1} - s_i)/s_i\big]_+$—inert on a decreasing term structure, active only where the model SSR rises with tenor. Fitting runs in two stages at uniform Gauss–Hermite resolution: a cold multi-start at n_{F}=7, then a warm Gauss–Newton refinement at n_{F}=17. Both are plotted in Figures 4–7 (θ@7, θ@17): the SSR is near-converged already at n_{F}=7, whereas the vol-of-vol—a higher moment—moves materially between the two, which is why every reported fit is read at n_{F}=17. The multi-basin freedom above means the cold start not infrequently settles in a poorer basin; where it does we re-seed rather than accept it, and flag the year in the *fit* column of Table 4: NDX 2019 is front-weighted (below), and SPX 2012/20/21/22 are re-seeded from a neighbouring good basin. SPX 2022 is the clearest case—the cold start rails ν_{F} at its floor and returns a 17.5% vol-of-vol, against 1.9% re-seeded with the SSR essentially unchanged. A re-seed is accepted only if it is lower-objective, non-railed, and stable under re-refinement.

Cross-regime fit. Each year is represented by a single fixed calendar date—the first June trading day, set by convention rather than chosen on fit quality—so the panel samples the regime, not the best-fitting day within it. Read at a *consistent* Gauss–Hermite resolution (n_{F}=17), the model reproduces both the realised SSR term structure and the VIX vol-of-vol across every regime from 2012 to 2024—the flattest (2012, roll-off 0.97) through the steepest (2024) to the COVID shock—to within the realised-SSR sampling noise on *both* readouts (Table 4, Figures 4–5). SSR RMS runs 0.7–8.1% and vol-of-vol RMS 1.1–6.5%, each inside the year's Newey–West HAC floor (its construction is Appendix F). The resolution is not incidental: the vol-of-vol is a high-moment readout that converges only near n_{F}≈17, so both functionals must be read at the same converged node count—at the production n_{F}=5 it is spuriously inflated, which is what earlier under-resolved fits mistook for a stress ceiling. Even the 2022–2023 grind is no exception: its realised SSR is normal (≈1.25–1.30) and the joint fit reaches SSR/vov RMS 2.5%/1.9% and 8.1%/5.4%. The two hardest vol-of-vol years (2023 melt-up, 2024; ≈5–6%) are the *lowest*-vol regimes, where the VIX vol-of-vol term structure carries a jagged short end (2023) or a non-monotone tail (2024) that lies within its own read noise; the smooth two-factor fits through it rather than chasing it—the same discipline as at the SSR short end (Figure 5). The vol-of-vol target also does not tax the SSR: on the 2022 grind a 25× increase in its weight moves the SSR RMS by under 0.1 point (2.53% → 2.57%)

while the vol-of-vol RMS *improves* ($1.92\% \to 1.72\%$), so the extra weight buys vol-of-vol accuracy at essentially no cost in SSR—the two readouts are near-orthogonal. For from-spot outputs the exact second-order leverage (17) is retained: the spot-conditional $\mathbb{E}[\nu \mid z{=}0] \approx 0.7$ (not 1) lifts the accumulated from-spot ATM level by $\sim$10% in vol.

Off-SPX generalisation (NDX). The model travels off SPX with *own-strip and realised targets only*, no vol-index options: NDX statics from its own SANOS marginals, the leverage from its realised SSR, and the vol-of-vol from the realised variance-of-variance of its own model-free variance-swap strip (the graded readout of Section 6; the reconstructed SPX $\sqrt{V}$ tracks the VIX). Across 2012–2024, once its thin-weekly short-dated corruption is cleaned (see the data note), NDX's realised SSR fits *within* its Newey–West noise floor in every regime (SSR RMS 1.6–6.5% against floors 7.5–15.5%). Correcting an earlier impression: those clean floors match SPX's (6–17%), so NDX's leverage is *not* meaningfully noisier off SPX—the apparent excess noise was the thin-weekly artifact. The genuine off-SPX limitation is the *vol-of-vol*: without a vol-index it is a *realised* ($\mathbb{P}$) reading, and—unlike the SSR's measure-invariant covariation (Appendix F)—its higher moments carry a $\mathbb{P}/\mathbb{Q}$ gap, so off-SPX the vol-of-vol is best read as an upper bound rather than a risk-neutral match. Cleaning the readout sharpens rather than softens the picture. We read it from a *cleaned constant-maturity* variance-swap strip at five tenors (30–180 d)—interpolating total variance to a fixed tenor exactly as VIX does, which removes the thin-expiry roll artifacts that corrupt the raw nearest-expiry readout (data note)—and fit the kernel only to the two liquid anchors (30/90 d), so the cleaned 45/120/180 d realised var-of-var is *predicted* out-of-sample in tenor. Most years then fit to a few percent (2012/16/17/22 to $\leq$2.1%, 2020 to 3.6%, 2019 to 5.6%); the residual concentrates in the three steep-decay melt-up years (2021, 2023, 2024), where the realised variance-of-variance falls from 30 to 90 days faster (ratio $\approx$0.4) than the two-factor kernel's floor ($\approx$0.58) permits—a 12–23% *shape* gap. This is a two-factor shape limit, *not* a case for a third factor (a third factor was tried, twice, and did not close it), and cleaning *removes* the earlier two-tenor-noise confound: the gap is now a reproducible property of the two-timescale decay, not target noise. One caveat belongs with that diagnosis, and it is a *measure* caveat rather than a noise one. The NDX target is a *realised* ($\mathbb{P}$) variance-of-variance, whereas the kernel's vol-of-vol is risk-neutral ($\mathbb{Q}$): forward implied volatility needs the transition density, not merely the marginals [19], and that density is a $\mathbb{Q}$ object. On SPX—the one underlying where both are observable—we measure the gap directly, pricing the VIX-implied $\mathbb{Q}$ vol-of-vol against a realised $\mathbb{P}$ var-of-var built from SPX's own strip by the identical constant-maturity construction. The ratio is not a level shift but a *tilt*, monotone in tenor: median $\mathbb{Q}/\mathbb{P} \approx 0.81$ at 25–40 d, crossing 1 near 60–90 d, and reaching $\approx$1.4 beyond 230 d (nine years, tight interquartile ranges). A $\mathbb{P}$ variance-of-variance therefore falls off with tenor markedly faster than its $\mathbb{Q}$ counterpart—the same direction as the shape gap above. We cannot verify this on NDX, since the absence of a liquid vol index is exactly why the target is realised there, and some of the wedge may be construction rather than premium (VIX-option implied volatility and the realised volatility of a constant-maturity variance series are not the same functional). The shape gap is therefore best read as an *upper bound* on the model's shortfall: part is the two-factor floor, and part is the measure in which the target is quoted. For 2019 the default joint weighting settles in a mis-converged basin (1wk SSR overshooting to 1.94, SSR RMS 10.2%); the front-weighted (SSR-first) fit escapes it, recovering 2.0% SSR and 5.6% vov at 1wk = 1.70 against the realised 1.68. More generally, where refitting the vol-of-vol to the cleaned anchors would pull the first-order leverage past its noise floor we keep the SSR-first fit, so the SSR stays within noise and the vol-of-vol reports its honest ceiling. Cross-sectionally NDX's realised vol-of-vol ($\sim$1.5) exceeds SPX's ($\sim$1.2)—tech is a jumpier vol process—and its marginal is modestly fat-tailed ($\sim$2–3$\times$ the data's excess kurtosis), a discretisation feature. Per-year term structures in Figures 6–7.

under.	regime	SSR RMS	VIX / vov RMS	fit
SPX	2012 flattest	3.7%	1.1%	re-seed
SPX	2016 steep	0.7%	2.0%	—
SPX	2017 steep/calm	1.7%	3.0%	—
SPX	2019 moderate	2.7%	2.1%	—
SPX	2020 COVID	2.5%	2.4%	re-seed
SPX	2021 high-flat	3.2%	2.0%	re-seed
SPX	2022 bear grind	2.5%	1.9%	re-seed
SPX	2023 melt-up	8.1%	5.4%	—
SPX	2024 steep low-vol	3.5%	6.5%	—
NDX	2012 flattest	3.1%	1.1%	—
NDX	2016 steep	2.4%	2.0%	—
NDX	2017 steep	6.3%	2.0%	robust-β
NDX	2019 moderate	2.0%	5.6%	front-wt.
NDX	2020 COVID	1.9%	3.6%	—
NDX	2021 melt-up	4.4%	17.1%	—
NDX	2022 bear	6.5%	2.1%	—
NDX	2023 melt-up	1.6%	11.9%	—
NDX	2024 low-vol	1.7%	23.3%	—

Table 4: Fit quality across regimes and underlyings at consistent n_{F}=17 (SPX: SSR+vov, the vov read from VIX; NDX: SSR+realised variance-of-variance, no VIX). SPX is one representative date per year, 2012–2024, spanning the flattest term structure (2012) to the steepest (2024) and the COVID shock; every regime fits within its realised-SSR Newey–West HAC floor on *both* readouts (SSR 0.7–8.1%, vov 1.1–6.5%; term structures in Figures 4–5). NDX is the same 9 years on own-strip/realised targets (no VIX), *both* readouts cleaned (data note) and at the same n_{F}=17; its SSR fits within its floor every year and its cleaned constant-maturity variance-of-variance fits to a few percent except the steep-decay melt-up years (2021, 2023, 2024; 12–23%), where the two-factor's floor on the 30→90-day vol-of-vol decay cannot match the realised (Figures 6–7). In the lowest-vol years (SPX 2017/21/24, NDX 2024) the realised 1-week SSR exceeds the diffusive bound 2 and is reported, not capped. The *fit* column records the per-year protocol: "—" the default OLS-β cold-start fit, "robust-β" a Huber target (NDX 2017), "front-wt." the front-weighted objective (NDX 2019), and "re-seed" a warm start from a neighbouring good basin (SPX 2012/20/21/22); see the fitting protocol above.

Smile dynamics: the level rides, the shape sticks. The same exact-β readout, taken across the whole at-the-money neighbourhood rather than at the level alone, places the calibrated model on the sticky map (the convention of Section 6) in two independent coordinates (Figure 9). The *level* rides with spot at the SSR, $\partial_{\ln S}\sigma_{\mathrm{ATM}} = \mathrm{SSR}\,\mathcal{S}$, with SSR running from 2.0 at one week down to 1.4 at three months. That is far from sticky-moneyness (SSR=0), between sticky-strike (1) and the local-vol limit (2), and on the Doeff–Kamal $H+\frac{3}{2}$ backbone through the belly. The *shape*, by contrast, barely moves. The skew's own response $\partial_{\ln S}\mathcal{S}$, normalised by the rate at which a strike-pinned smile would swing it (twice the curvature), is only 0.01–0.02. The skew is essentially *frozen in moneyness* (sticky-moneyness *in shape*), with a small, realistic steepening on sell-offs. So the model reproduces the textbook equity picture—the smile *shape* translates with spot while its *level* lifts on sell-offs (Figure 9b: the one-month smile shifts up nearly in parallel on a 1% drop, unlike

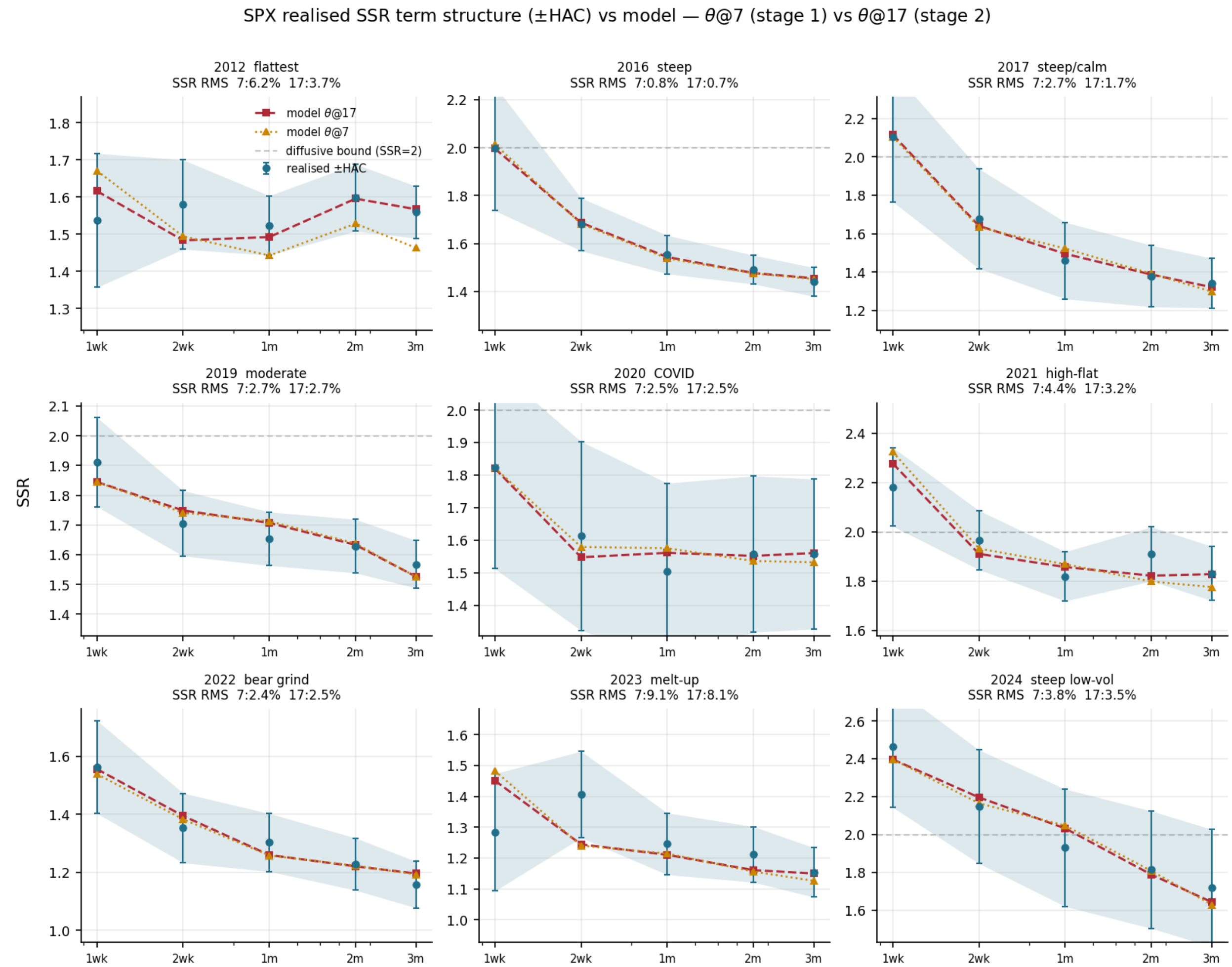

Figure 4: Realised SSR term structure vs. model, one representative date per year 2012–2024, at consistent n_{F}=17. Markers with error bars and the shaded band: the realised skew-stickiness (a 250-day $\mathbb{P}$ regression) with its Newey–West HAC ± 1 SE. Dashed/dotted: the model SSR at the fitted θ (θ@17, θ@7). The model sits inside the sampling band from the flattest regime (2012) through the steepest (2024) and COVID (2020). The plain dashed line marks the diffusive $H=\frac{1}{2}$ short-end bound SSR=2; in the lowest-vol years (2017/21/24) the *realised* 1-week SSR itself runs above it (≈2.1–2.5)—we report the realised rather than capping at 2, and the discrete kernel tracks it (these short-end points sit beyond the strict $H=\frac{1}{2}$ scope, a noted limitation).

the flat sticky-moneyness or the tilted sticky-strike responses). This is also why the SSR-consistent delta is the right hedge: with the shape sticky and only the level riding, the whole hedging problem collapses to folding in that level move—the SSR—which is exactly what (26) does.

The frozen shape, however, is as much a *structural constraint* as an empirical match. By translation invariance (Section 3) the conditional smile depends on the volatility regime alone, not on the spot level. So the shape can respond to a spot move only *indirectly*, through the leverage-induced regime shift: a down move tilts the regime toward higher, steeper-skew vol (the small sell-off steepening above). A *deterministic, spot-level-dependent* shape—a smile intrinsically steeper at lower index levels, beyond that stochastic-regime channel—is outside the model's reach:

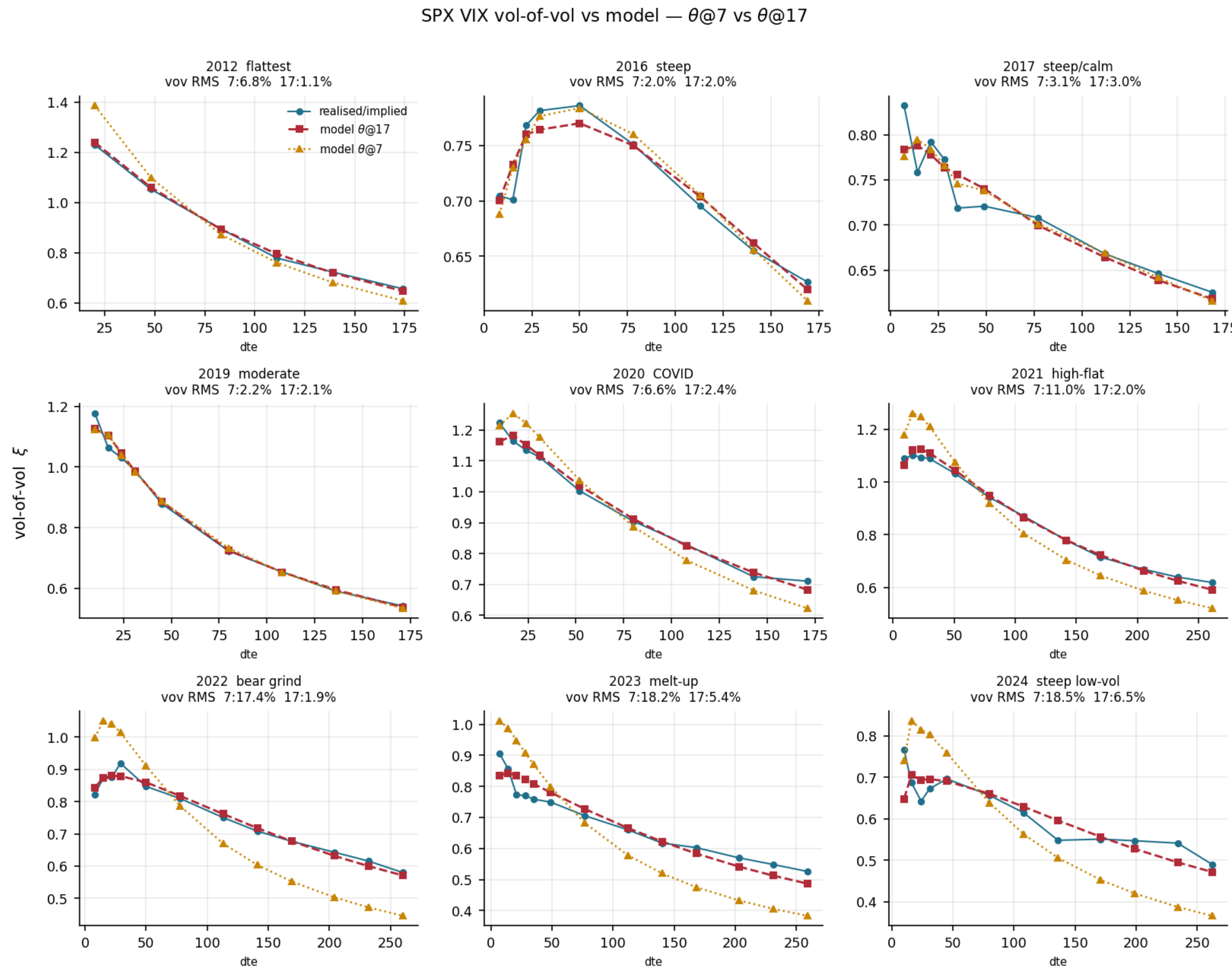

Figure 5: VIX vol-of-vol term structure vs. model, same dates. Markers: the VIX-implied ATM vol-of-vol ξ across VIX-option maturities—a *single-day* $\mathbb{Q}$ read, so (unlike the SSR) it carries no sampling band. Dashed: model. The fit is tight in most regimes; the visible residual in the two lowest-vol years (2023, 2024) is a jagged short end / non-monotone tail in the *data*—within its read noise—that the smooth two-factor declines to chase.

the same invariance that makes the SSR an exact closed form forbids it. So where the data carry such second-order, level-dependent skew dynamics, the "shape sticks" finding is a property of the construction as much as of the market.

Hedging: the SSR-consistent delta. We test the "better delta" claim by historical replay. We delta-hedge a rolling one-month ATM SPX option once per day over the one-parameter family $\Delta(R) = \Delta_{\mathrm{BS}} + \mathrm{Vega} \cdot R\mathcal{S}/S$ (26), which spans the standard deltas: $R{=}0$ is Black (sticky-strike), $R{=}-1$ is sticky-delta, and R equal to the SSR is the minimum-variance delta. Black enters as the *no-adjustment reference* that sets the scale, not as a competitor: beating it is the *expected* minimum-variance-delta effect on an equity index [23, 34], and the benchmark a desk actually hedges against is the Hull–White delta of the next paragraph. The substantive check is where the curve bottoms—the one-day hedging-P&L variance is minimised at $R^\star$ equal to the *realised*

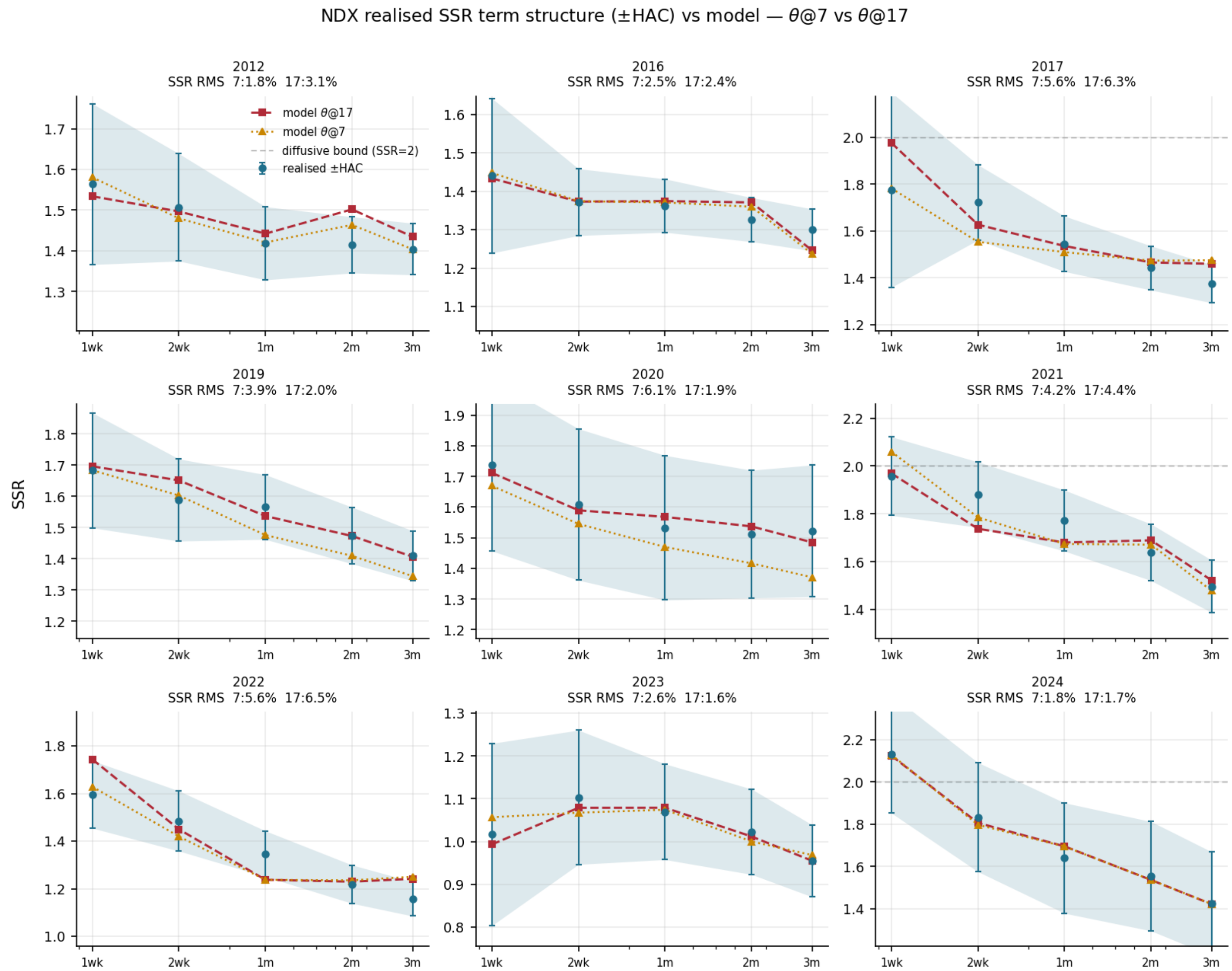

Figure 6: Off-SPX (NDX): realised SSR term structure vs. model, the same 9 years, own targets, n_{F}=17. Markers/band: realised skew-stickiness ±Newey–West HAC, after the thin-weekly cleaning (data note). Dashed: model. The model tracks within the sampling band in every regime; the clean floors (7.5–15.5%) match SPX's. The plain dashed line is the diffusive $H=\frac{1}{2}$ bound SSR=2; NDX 2024's realised 1-week SSR runs above it (reported, not capped).

SSR (Figure 10, Table 5). Across 2016–2021, $R^\star \approx 1.5$–1.8, tracking the realised SSR (1.5–1.8); hedging at the model's structural R=1.6 cuts the realised hedging variance 30–50% against that reference. The edge persists through the 2022–2023 grind—realised SSR ≈1.3, reduction 15–37% (Table 5)—so the SSR-consistent delta helps in the grind as well, its spot–vol comovement being normal rather than degenerate. Sticky-delta ($R=-1$) is *worse* than Black: it moves the delta with the vol–spot comovement the wrong way, the sign check the family passes.

Benchmark: the Hull–White minimum-variance delta. The competitor is the industry MV delta of Hull–White [34]—a trailing regression of implied-vol changes on returns, i.e. the *trailing realised*-SSR delta—against which we run the model's forward-looking, *option-implied* SSR (the skew-decay $H+\frac{3}{2}$ readout of Section 6.4, independent of the realised estimate, so the test is not circular). Out of sample on SPX (50/50 train/test, clean spot), the model delta carries $\mathbf{0.76}\times$ the

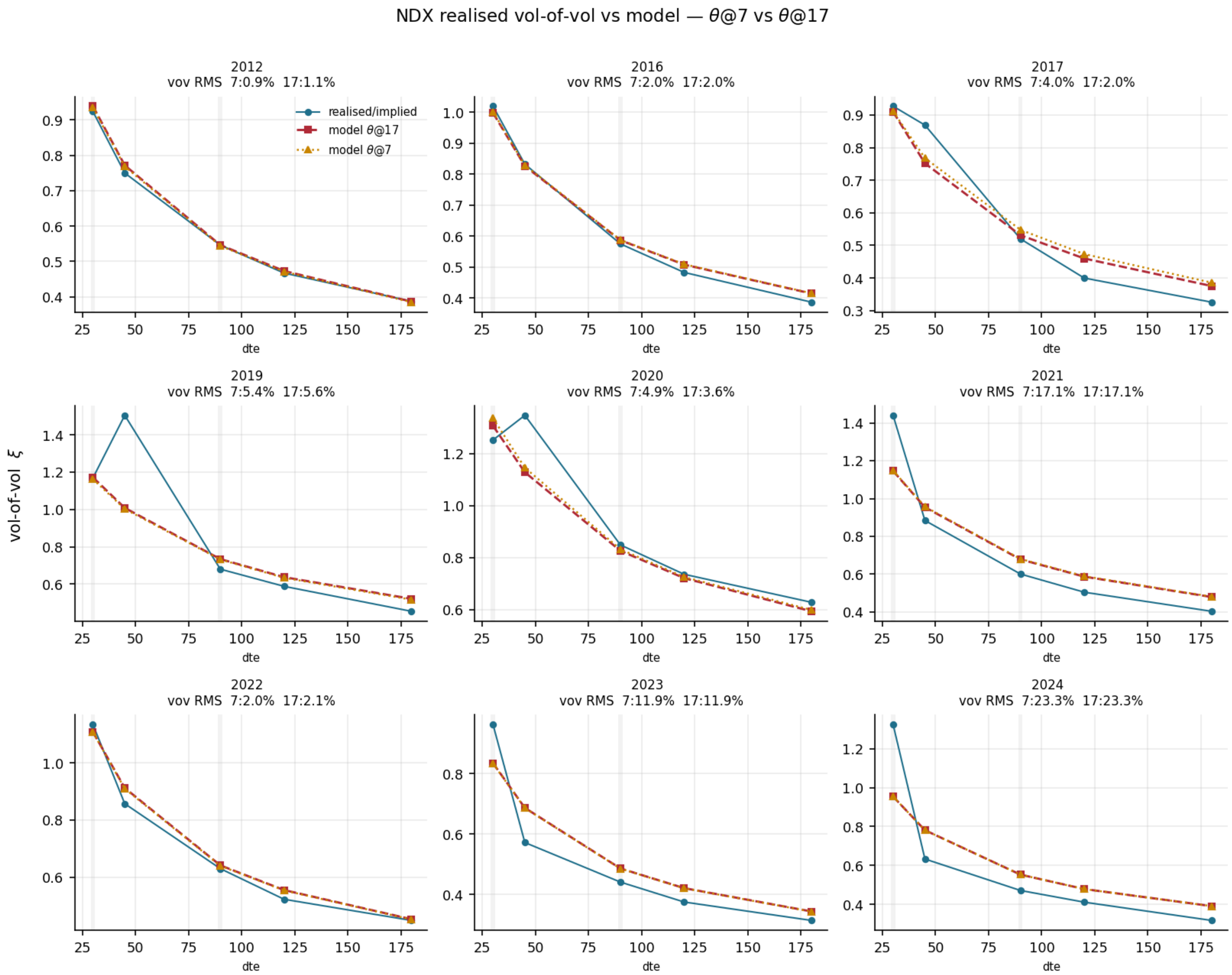

Figure 7: Off-SPX (NDX): *cleaned constant-maturity* realised variance-of-variance at five tenors (30–180 d, $\mathbb{P}$) vs. model vol-of-vol, n_{F}=17. The kernel is fit to the two liquid anchors (30/90 d, shaded); the 45/120/180 d points are *predicted* out-of-sample in tenor. Most years fit to a few percent; the steep-decay melt-up years (2021, 2023, 2024) show the realised var-of-var falling from 30 to 90 days faster than the two-factor floor permits—a 12–23% shape gap. The COVID-era 45-day read (2019/20) carries residual roll noise the smooth kernel does not chase (an interpolation artifact where NDX's expiry ladder thins past the front weeklies, not a vol feature). Titles quote the 30/90 fit RMS.

Hull–White delta's hedging variance over 2015–2019 (Diebold–Mariano p=0.023) and 0.89× over the full 2015–2023 (p=0.16); the result is robust to the trailing window and to the cross-strike Hull–White form (correlation 0.98 with the single-strike version) (Figure 11). Two qualifications keep this honest. *First, neither delta forecasts comovement:* the option-implied and trailing-realised SSR both correlate ≈0 with next-period realised comovement (encompassing test inconclusive), which is simply unforecastable at this horizon. *Second, the edge is stability, not prediction.* That a *stable* delta beats the *adaptive* Hull–White estimator is the substantive point: SPX spot–vol comovement mean-reverts fast enough that the trailing regression over-adapts, chasing realised noise that does not persist (Figure 11a). But this stability is not special to the option-implied R—a *constant* R

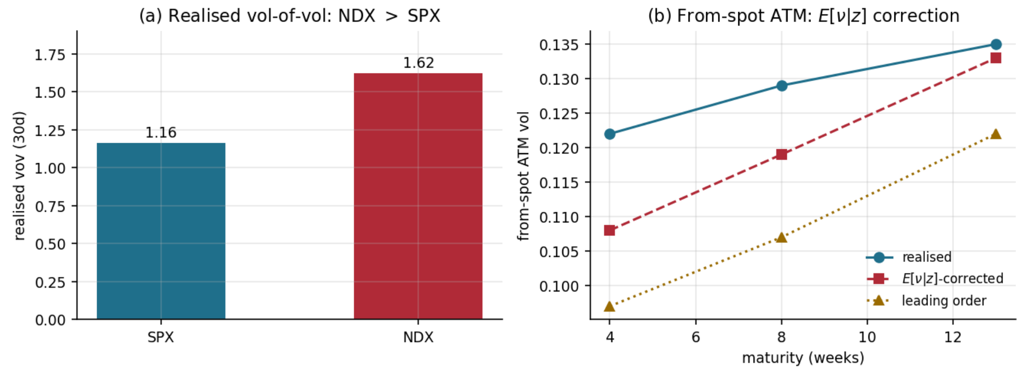

Figure 8: Off-SPX (NDX): the two readouts *not* carried by Table 4. **(a)** NDX's realised vol-of-vol exceeds SPX's—tech is a jumpier vol process. **(b)** The second-order $\mathbb{E}[\nu \mid z]$ leverage correction lifts the accumulated from-spot ATM level toward the realised. (The per-year SSR/vov fit RMS previously shown here duplicated Table 4; the term structures are Figures 4–7.)

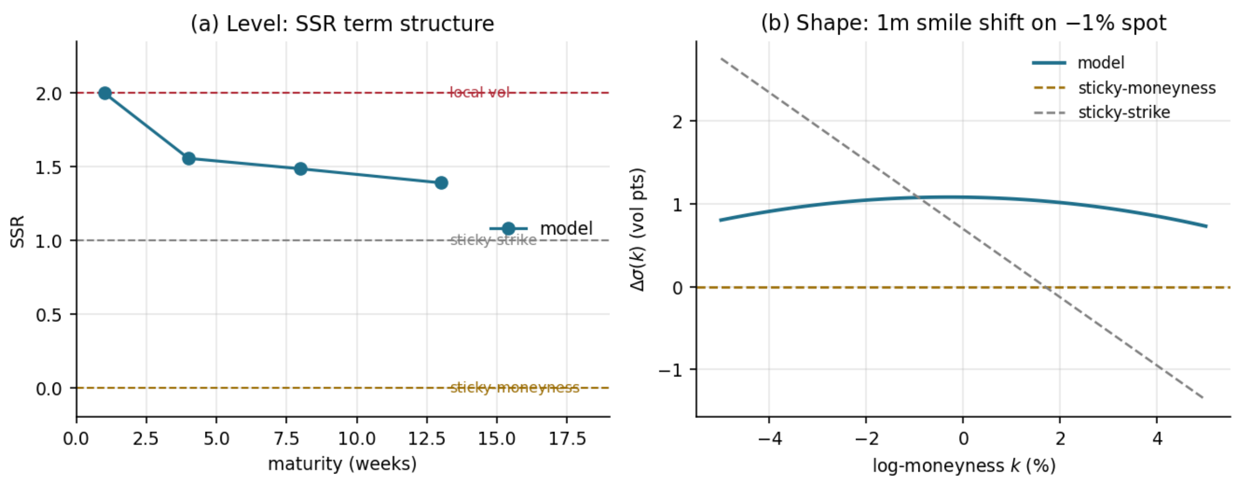

Figure 9: Where the calibrated model sits on the sticky map (SPX, the exact conditional-smile response to a spot move, decomposed by moment). **(a)** *Level*: the SSR term structure runs $2.0 \to 1.4$—between sticky-strike (1) and local vol (2), far above sticky-moneyness (0). **(b)** *Shape*: the one-month smile's response to a -1% spot move is a near-parallel upward shift ($\sim +1$ vol point), close to the flat sticky-moneyness line and unlike the tilted sticky-strike one—the shape is frozen in moneyness while the level rides.

does about as well ($0.80/0.85\times$). So on SPX the model's value is not that its implied R out-hedges a fixed number; it is that the smile alone *supplies* the right stable level—no realised history, no rolling regression—which a constant cannot do on an asset or regime where that level is not already known. The edge is largest in normal regimes (SSR $\approx \frac{3}{2}$) and survives transaction costs at SPX liquidity.

Scope: ATM dynamics; full-smile dynamics as future work. The statics match the full vanilla smile, but the *dynamic* layer is calibrated to *at-the-money* observables—the SSR and the

year	days	realised SSR	$R^\star$	var. reduction vs Black
2016	251	1.56	1.75	47%
2017	250	1.46	1.50	30%
2019	251	1.66	1.80	47%
2021	251	1.82	1.80	50%
2022	250	1.30	1.55	37%
2023	249	1.24	1.05	15%

Table 5: SSR-consistent hedging (one-month ATM SPX, daily rehedge). The variance-minimising delta parameter $R^\star$ tracks the realised SSR; hedging at the model's structural R=1.6 reduces the one-day hedging variance 30–50% versus Black in normal regimes, and still 15–37% through the 2022–2023 grind (below the rule), whose realised SSR is itself normal. Sticky-delta ($R = -1$, not shown) *increases* it by 38–44%.

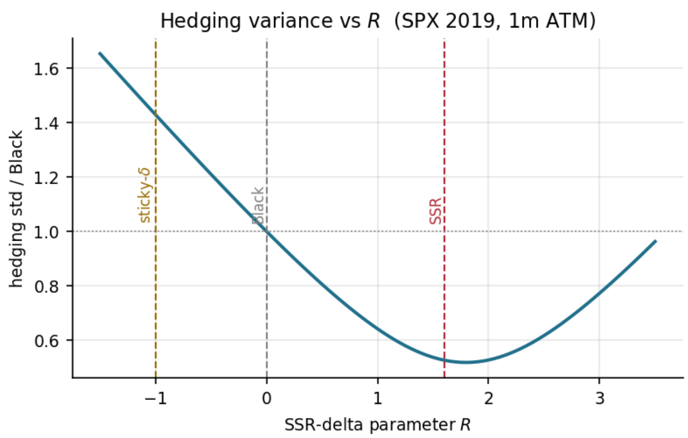

Figure 10: Hedging replay (one-month ATM SPX, daily rehedge). The one-day hedging-P&L variance, normalised to the no-adjustment reference, as a function of the SSR-delta parameter R, for 2019: a clean U minimised near the *realised* SSR, with Black (R=0) and sticky-delta ($R = -1$) as the reference points either side—the latter worse, moving the delta with the comovement the wrong way. The per-year variance reductions are Table 5.

VIX ATM vol-of-vol. Two full-smile extensions are left to a sequel. First, the VIX *smile*—its skew and tails, not only the ATM level—is a richer risk-neutral reading of the vol-of-vol *shape*; the atomic VIX law (Sec. 6) prices it across strikes with no new machinery, tightening the vol-of-vol identification, though—being vol-of-vol—it is *complementary to, not a substitute for*, the leverage that carries the SSR. Second, the *forward-start* smile skew is the clean risk-neutral observable of the spot–vol leverage. It pins the slow leverage λ_S that vol-of-vol leaves soft (Appendix D), and it would supersede the realised-SSR (physical-measure) proxy used here. But it requires traded forward-start / cliquet quotes, which are over-the-counter and absent from listed-option feeds (ORATS included), so this route is contingent on data. Together these carry the calibration from ATM- to full-smile dynamics, of interest for cliquet and forward-volatility pricing.

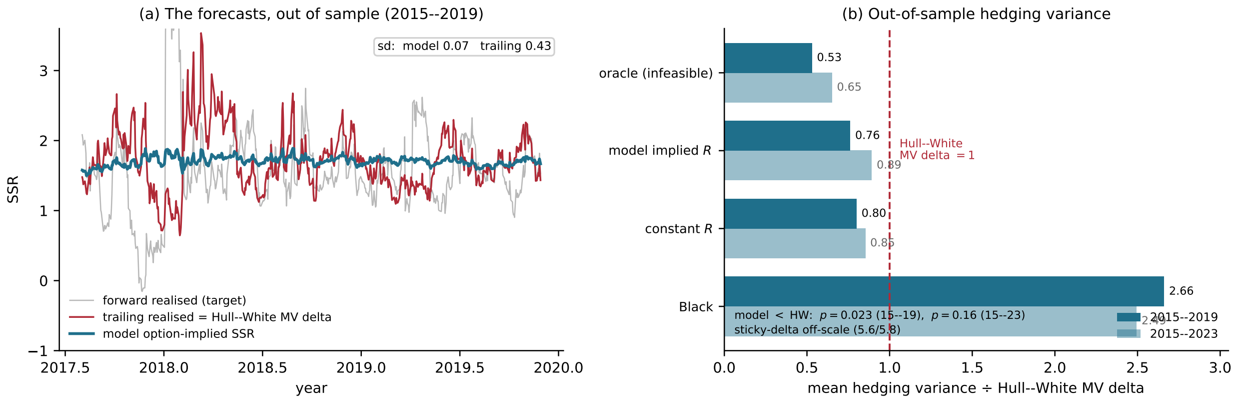

Figure 11: Benchmark: the model delta vs the Hull–White minimum-variance delta, out of sample (SPX, 50/50 train/test, clean spot). **(a)** The forecasts over the test window: the model's option-implied SSR (stable, sd 0.07) against the trailing-realised SSR that *is* the Hull–White MV delta (sd 0.43) and the forward-realised target. The stability contrast—not a better forecast—is what drives the edge. **(b)** Mean out-of-sample hedging variance per delta, normalised to the Hull–White MV delta (= 1): the model implied-R delta (0.76/0.89 for 2015–2019/2015–2023) and a constant-R delta (0.80/0.85) both sit below it, Black far above, sticky-delta off-scale. The model beats Hull–White with p=0.023 on 2015–2019.

8 Discussion and conclusion

With the construction and the empirical results in hand, we can be precise about what SANOS-Evolve buys over classical SLV, which already matches vanilla marginals and produces smile dynamics.

The differentiator, previewed in Section 1, is the SSR. The documented rigidity of the calibrated stochastic-volatility family [27, 25] is established for the *pure* SV class, but it transfers to classical SLV. The leverage is *consumed* matching the vanilla smile, so the residual spot–vol dynamics—and hence the SSR—falls to the stochastic-volatility backbone [24, 15]. An SLV on a standard backbone inherits the same band. SANOS-Evolve is itself a discrete SLV, so its escape cannot come from the leverage. It comes from the *backbone*: a discrete regime-transition kernel whose two-timescale innovation leverage shapes the SSR term structure directly, while the SANOS LP holds the marginals fixed. The calibration of Section 7 realises this on the empirical SPX term structure.

The differences are *not* approximation power—both are universal at the marginal (SLV through a nonparametric leverage function; SANOS-Evolve through W_1-dense Gaussian-mixture marginals and a sufficiently rich kernel). They are *where the structure sits* and *whether the dynamics are controlled*. Classical SLV sits on the other side of each contribution of Section 1. Its SSR is whatever the stochastic-volatility backbone produces, not a target. Its leverage is a fixed point of the calibration, so re-tuning the dynamics re-triggers it, and it can fail outright when vol-of-vol is large—the leverage $\sigma^2_{LV} = \sigma^2_{\text{Dupire}}/\mathbb{E}[V \mid S]$ diverging. The admissible set here is never empty (Strassen, Theorem 1). Its Monte-Carlo particle noise (worst in the wings, seed-dependent day to day) or PDE grid error sits inside the hedge ratios, where our mixture pricing is deterministic. And its dynamics live in a nonparametric leverage surface, against our nine interpretable knobs and latent regimes.

Universal statics, parsimonious dynamics. The two layers make opposite choices, on purpose. The static layer is universal: W_1-dense Gaussian mixtures (Appendix C) fit *any* arbitrage-free smile exactly—a structural guarantee, not a fitted approximation—so the smile match is never the binding constraint. The dynamic layer is parsimonious. The small propagation residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$ is a modelling choice, not a wall: admissibility is exact regardless, because the leverage overlay carries the statics and recompression re-locks the martingale (Section 3.3). A few knobs pin the dynamics to interpretable regimes where an unconstrained kernel would leave them underdetermined. What lets universality and parsimony coexist is translation-invariance. Exact marginal match, few interpretable parameters, and exact closed-form closure are jointly unsatisfiable for a *state-dependent* kernel. They hold together here because the translation-invariant kernel has exact Gaussian-mixture closure on the parsimonious generator.

Costs and limitations. The model is discrete, so continuously-monitored barriers need substepping. Recompression keeps the component budget finite (a computational cost, not an approximation limit). The one first-order approximation is the smooth-leverage collocation on the statics, with the dynamics staying exact by translation-invariance. And the framework is empirically unproven relative to battle-tested SLV. Against *frontier* SLV (multi-factor, rough, or neural-SDE [17]), which also fit dynamics flexibly, the edge narrows to the closed-form/deterministic/interpretable combination rather than dynamics control per se. Three further caveats: the empirical edge is bounded by *scope*, not regime—the SSR+vov fits sit within the realised-SSR noise floor across 2012–2024 (Table 4), but the diffusive $H=\frac{1}{2}$ kernel sets the sub-month SSR level and the smooth two-factor leaves a small vol-of-vol residual in the lowest-vol years; the two-timescale split is itself the least-identified block (Appendix D); and the static layer rests on SANOS [1], at this writing a preprint. Table 6 is the precise accounting: what is exact, what is approximate, and where each approximation sits.

Outlook: intraday and 0DTE. Same-day-expiry (“0DTE”) options are now a majority of S&P 500 option volume, and the construction ports there by *scale invariance.* The marginal mixtures, the kernel, and recompression and interpolation (Secs. 4.7, 4.8) are clock-agnostic. Rescaling the two Ornstein–Uhlenbeck timescales—fast κ_{F} in minutes, slow κ_{S} in hours—re-targets the model at the 0DTE surface. Its natural dynamic target is the *intraday* SSR: by the same $D(\kappa T)$ damping that makes VIX slow-dominated, a minute/hour vol factor is averaged out of any multi-day option and is visible only there. A genuine intraday model needs two localized additions—a time-of-day seasonality $\phi(u)$ and a terminal/pin refinement as $T \to 0$—plus synchronized intraday NBBO data, the empirical demands a full ultra-short surface study makes plain [29]. We leave the empirical 0DTE study to a dedicated sequel.

Summary. SANOS-Evolve keeps SANOS's arbitrage-free marginals and evolves them into a discrete stochastic-local-volatility model by fitting a martingale two-timescale kernel: the marginals stay strictly arbitrage-free while the kernel supplies the smile dynamics local volatility gets wrong, with the SSR an exact closed-form realised covariance and the vol-of-vol identified—not jointly fitted—from a forward-variance readout. Everything is closed-form, no-arbitrage is a theorem with Strassen existence, and the desk outputs—forward densities and SSR-consistent deltas—are functionals of one calibrated object. The empirical study validates the machinery on synthetic ground truth and calibrates it across real SPX regimes, with a hedging replay that isolates a modest, stability-driven edge over the industry minimum-variance delta. What remains is the broader multi-moneyness, multi-tenor study and the intraday 0DTE sequel outlined above.

status	object	note
exact (closed form)	the translation-invariant kernel and per-fibre lock (§4); the realised-covariance SSR (23); the ATM Taylor coefficients (App. E); the mixture cumulants; static no-arbitrage (Thm. 1, Strassen)	independent of every dynamic approximation; z-invariance is what makes the SSR exact, Monte-Carlo-confirmed ($\lvert\text{diff}\rvert \approx 0.005$)
leverage collocation (check first)	the smooth local-vol leverage $\sigma_{\text{LV}}(z)$ collocated at the component means (§4.6)	moves the statics *level*, not the SSR mechanism (which stays z-exact); single-step benign, but a $\sim$10% multi-month vol-level bias if the exact $\mathbb{E}[\nu \mid z]$ projection (17) is dropped (§4.6)
controlled (convergent)	recompression to $n_{\text{keep}} \approx 24$ (§4.7); the discrete Dupire under-reads at narrow marginals ($\sim$50–85 bp short-T)	accuracy dials—central-difference Dupire, finer grids, tail-clamped leverage
leading-order (cross-check)	the cumulant-to-IV route (§6.4, $\sim$8–9% off)	*not* the primary readout; the exact coefficients (App. E) supersede it
structural limit	short-T diffusive SSR$\to 2$ ($H = \frac{1}{2}$) vs SPX's rough ≈ 1.5	a modelling gap, not numerical error; fix = a rough multi-factor extension
structural limit	spot-level-*independent* smile *shape*: the conditional smile depends on the vol regime, not z (§3)	shape responds to spot only via the stochastic-regime channel; deterministic level-dependent skew dynamics are out of reach—the flip side of translation-invariance
out of scope	the wings / Lee moment-index regime	a finite Gaussian mixture is structurally thin-tailed there; ATM by design

Table 6: What is exact and what is approximate. The contrast with a state-dependent (spot-level) kernel is the point: there the quantity to check first is a measure-transport collapse *inside the dynamics*; here the only first-order approximation is a smooth-leverage collocation *on the statics*, the dynamics left exact by translation-invariance.

Acknowledgements

I would like to thank my fellow traders for helpful discussions during the preparation of this work.

A The SANOS marginal-calibration LP

For completeness, Algorithm 2 states the SANOS static-layer calibration in this paper's notation; the original is [1]. It fits, per maturity, a weights-only Gaussian mixture whose call surface lands within bid–ask and is strictly free of butterfly and calendar arbitrage (positivity → valid density; forward constraint → martingale; convex-order inequality → calendar). The problem is convex—a linear program for a piecewise-linear smoothing penalty $\mathcal{R}$, quadratic for a quadratic one—and feasible within the quote bands by construction. This is the admissible-marginal layer $\{\mu_j\}$ on which Theorem 1 and its convex-order precondition rest.

Algorithm 2 SANOS marginal-calibration LP at maturity T_j (this paper's notation; after [1]).

Require: strikes $\{K_{j,m}\}$ with bid/ask $[B_{j,m}, A_{j,m}]$, mids $C^{\text{mid}}_{j,m}$, weights $\omega_{j,m}$; forward F_j; fixed anchors $\{a_i\}_{i=1}^N$ in $z = \log(S/F_j)$; bandwidth h; previous-maturity fit q_{j-1}

Ensure: weights $q_j \geq 0$ with $\rho_{S,T_j}(z) = \sum_i q_j^i \,\mathcal{N}(z; a_i, h^2)$, strictly arbitrage-free

1: Build the linear pricing map $(\Pi_j)_{m,i} = \mathrm{BS}\big(F_j e^{a_i + h^2/2},\, K_{j,m},\, h\big)$, the price of basis component i at strike $K_{j,m}$.

2: Solve the linear program over $(q_j, t) \geq 0$:

$$
\begin{aligned}
\min \quad & \sum_m \omega_{j,m}\, t_m \;+\; \lambda_h\, \mathcal{R}(q_j) \\
\text{s.t.} \quad & -t_m \leq (\Pi_j q_j)_m - C^{\text{mid}}_{j,m} \leq t_m \qquad (L^1 \text{ fit to mids}) \\
& B_{j,m} \leq (\Pi_j q_j)_m \leq A_{j,m} \qquad (\text{hard bid–ask box}) \\
& \mathbf{1}^\top q_j = 1, \quad \textstyle\sum_i q_j^i\, e^{a_i + h^2/2} = 1 \qquad (\text{normalisation; forward}) \\
& \widehat{C}_j(K_g) \geq \widehat{C}_{j-1}(K_g) \;\; \forall K_g \text{ on a dense grid} \qquad (\text{calendar / convex order})
\end{aligned}
$$

3: **return** q_j; the CDF is $G_{S,T_j}(K) = \sum_i q_j^i\, \Phi\big((\log(K/F_j) - a_i)/h\big)$.

B Proof sketch of Theorem 1

Strike no-arbitrage. For each j, μ_j is a probability measure with $\mathbb{E}_{\mu_j}[S] = F_j$; hence $C_j(K) = D_j \int (s-K)^+ \mu_j(\,\mathrm{d}s)$ is convex and non-increasing in K with the correct bounds, which is equivalent to absence of butterfly arbitrage (Breeden–Litzenberger).

Martingale and marginals. Admissibility gives $\mathbb{E}[e^{Z_{j+1}} \mid X_j] = e^{Z_j}$, i.e. the forward-normalised price is a martingale, and $\mu_j \mathcal{K}_j = \mu_{j+1}$ propagates the law, so the chain has one-dimensional marginals exactly $\{\mu_j\}$ and the discounted underlying is a martingale.

Calendar no-arbitrage. A martingale transition implies, by conditional Jensen, that the forward-normalised marginals are increasing in convex order, $\tilde{\mu}_j \preceq_{\text{cx}} \tilde{\mu}_{j+1}$, which is equivalent to absence of calendar arbitrage for the call surface.

Existence. By Strassen's theorem [20, 21], a martingale kernel with $\mu_j \mathcal{K}_j = \mu_{j+1}$ exists iff $\tilde{\mu}_j \preceq_{\text{cx}} \tilde{\mu}_{j+1}$. SANOS enforces this convex order across maturities, so $\mathcal{A}_j \neq \emptyset$. □

C Wasserstein-1 universality: marginals, dynamics, and kernels

That the mixture layer is *effectively nonparametric*—the expressiveness it gives the statics (Section 1)—rests on a denseness fact, which we make precise here *in the topology that governs option prices*. Gaussian mixtures approximate any arbitrage-free marginal, and any continuous admissible kernel, to arbitrary accuracy, with the approximant exactly forward-matched.

The pricing topology is W_1, not weak.* Work in the spot $s = e^z$. The observables are the undiscounted calls $\Lambda_k(\rho) = \langle \rho, (s-k)^+ \rangle$, whose payoff is 1-Lipschitz in s, together with the forward $\langle \rho, s \rangle$. The coarsest topology making all of these continuous is the Wasserstein-1 topology on $\mathcal{P}_1(\mathbb{R}_+)$, the laws of finite first spot-moment; by Kantorovich–Rubinstein and (20),

$$
\sup_k \big|\Lambda_k(\mu) - \Lambda_k(\nu)\big| \;\leq\; W_1(\mu, \nu) = \int_0^\infty |G_\mu(s) - G_\nu(s)|\, \mathrm{d}s = \big\|C_\mu - C_\nu\big\|_{\dot{W}^{1,1}},
$$

the aggregate digital mispricing—*exactly the band-loss the calibration minimises* (Section 5). Weak* convergence is strictly too weak here. The call payoff $(s-k)^+$ and the forward s are continuous but *unbounded*, hence not weak* test functions. So mass escaping to the tail moves prices while the marginal is held fixed against the bounded-continuous C_b. Wasserstein-1 is precisely weak* *plus* the uniformly-integrable first moment that pins the forward and the option values. The martingale constraint is the single linear condition $\langle\rho, s\rangle = F$.

Lemma 2 (Marginal universality in W_1)**.** *For any arbitrage-free marginal $\rho \in \mathcal{P}_1\big((0,\infty)\big)$ with forward $\langle\rho, s\rangle = F$ and any $\varepsilon > 0$, there is a finite Gaussian mixture η (lognormal in s, Gaussian in $z = \log s$) with $\langle\eta, s\rangle = F$* exactly *and $W_1(\eta, \rho) < \varepsilon$.*

Proof. Quantise, mean-preserving. Because $\langle\rho, s\rangle < \infty$, pick M with $\int_{s>M} s\,\rho < \varepsilon/4$ and partition $[0, M]$ into cells $C_1, \dots, C_N$; replace ρ on each cell, and on the tail (M, ∞), by an atom at its conditional *barycentre* $s_j = \mathbb{E}[S \mid C_j]$ with mass $w_j = \rho(C_j)$. The atomic η_{a} then has $\langle\eta_{\mathrm{a}}, s\rangle = F$ exactly, and—moving each unit of mass no further than its own cell, the tail cell's cost being $\le 2\int_{s>M} s\,\rho$—

$$W_1(\eta_{\mathrm{a}}, \rho) \;\le\; \max_{j\le N} \mathrm{diam}(C_j) \;+\; 2\int_{s>M} s\,\rho(\,\mathrm{d}s) \;\xrightarrow[\text{mesh}\to 0,\ M\to\infty]{}\; 0.$$

Mollify, mean-preserving. Replace each δ_{s_j} by a lognormal L_j with mean $\mathbb{E}[L_j] = s_j$ and log-variance σ^2 (a Gaussian in z); the forward stays F exactly, and by convexity of W_1 under mixing,

$$W_1(\eta, \eta_{\mathrm{a}}) \;\le\; \sum_j w_j\,\mathbb{E}|L_j - s_j| \;\le\; \Big(\textstyle\sum_j w_j s_j\Big)\sqrt{e^{\sigma^2} - 1} \;=\; F\sqrt{e^{\sigma^2} - 1} \;\xrightarrow[\sigma\to 0]{}\; 0,$$

using $\mathbb{E}|L_j - s_j| \le s_j\sqrt{e^{\sigma^2} - 1}$ (Cauchy–Schwarz). Choosing the mesh and M, then σ, small enough gives $W_1(\eta, \rho) < \varepsilon$ with the forward matched throughout. $\square$

Lemma 2 is the *statics*: the mixture layer reaches any arbitrage-free marginal. What the construction asks of the *kernel* is weaker—only the translation-invariant dynamics, because the leverage (Section 4.6) supplies the marginal match. For those, the mixture kernel is again dense.

Proposition 3 (Dynamic universality within the translation-invariant class)**.** *Let $\mathcal{K}$ be any translation-invariant martingale kernel—its conditional return law a function of a latent Markov volatility state, not of the spot level. For any $\varepsilon > 0$ there is a Gauss–Hermite Gaussian-mixture kernel of the form* (6)*, with enough carried factors and nodes, that is itself translation-invariant and per-fibre martingale* (13) *and lies within $W_1 < \varepsilon$ of $\mathcal{K}$ on each fibre—so it reproduces $\mathcal{K}$'s dynamic readouts, the SSR term structure and the forward-variance dynamics, to within ε.*

Proof. Fix a fibre and write the one-step conditional return $R = \Delta z$. Translation invariance means the conditional law $\mathrm{Law}(R \mid V{=}v) = q_v$ depends only on the latent volatility state V (a Markov process), not on the spot, with the per-state martingale lock $\int e^r q_v(\,\mathrm{d}r) = 1$; the map $v \mapsto q_v$ is W_1-continuous, as for the smooth Ornstein–Uhlenbeck volatility. Write the kernel on the fibre as the latent mixture $\mathcal{K}(z, \cdot) = \int q_v\,\pi_z(\,\mathrm{d}v)$, with π_z the law of V there. Three bounds compose.

(i) Increment, per state. Each $q_v \in \mathcal{P}_1$ has unit e^r-mean, so Lemma 2 applied to e^R yields a finite mean-preserving Gaussian mixture η_v with $\int e^r \eta_v(\,\mathrm{d}r) = 1$ and

$$W_1(\eta_v, q_v) < \varepsilon_1 \qquad \text{for every } v.$$

(ii) Latent law. Each Ornstein–Uhlenbeck factor carries a Gaussian weight; its n-node Gauss–Hermite quadrature $\hat\pi_n$ integrates every polynomial of degree $\le 2n{-}1$ exactly, so $W_1(\hat\pi_n, \pi) \to 0$ as $n \to \infty$, the carried transitions $P_{\mathsf{F}}, P_{\mathsf{S}}$ being the quadrature of the mean-reverting semigroup. With $v \mapsto q_v$ L-Lipschitz in W_1,

$$W_1\Big(\textstyle\int q_v\,\hat\pi_{n,z}(\,\mathrm{d}v),\ \int q_v\,\pi_z(\,\mathrm{d}v)\Big) \;\le\; L\,W_1(\hat\pi_{n,z}, \pi_z) \;=:\; \varepsilon_2(n) \;\xrightarrow[n\to\infty]{}\; 0.$$

(iii) Compose. The approximant $\hat{\mathcal{K}}(z,\cdot) = \int \eta_v \, \hat{\pi}_{n,z}(\,\mathrm{d}v)$ has the form (6). Convexity of W_1 under mixing gives, uniformly in z by translation invariance,

$$W_1\big(\hat{\mathcal{K}}(z,\cdot), \mathcal{K}(z,\cdot)\big) \;\le\; \int W_1(\eta_v, q_v)\, \hat{\pi}_{n,z}(\,\mathrm{d}v) \;+\; W_1\Big(\textstyle\int q_v \, \hat{\pi}_{n,z}, \int q_v \, \pi_z\Big) \;<\; \varepsilon_1 + \varepsilon_2(n).$$

Take $\varepsilon_1 < \varepsilon/2$, then n with $\varepsilon_2(n) < \varepsilon/2$. Every η_v has unit e^r-mean, so $\hat{\mathcal{K}}$ meets the per-fibre lock (13) exactly and, of the form (6), is translation-invariant and—fused with the leverage—admissible.

Readouts. The mixture components carry uniformly bounded log-variance, so they are lognormal with uniformly bounded parameters and hence have uniformly bounded moments of every order; the resulting uniform integrability of the squares (not mere boundedness of the second moment) upgrades the W_1 bound to W_2, under which the realised SSR (23)—a ratio of one-step return/volatility covariances with denominator bounded below—and the forward-variance readout are continuous. Hence $\mathrm{SSR}_T(\hat{\mathcal{K}}) \to \mathrm{SSR}_T(\mathcal{K})$ and the forward-variance dynamics converge, each to $O(\varepsilon)$. □

The construction uses this proposition—and a parsimonious two-factor instance of it, enough for the SPX SSR term structure (Section 7)—with the leverage, not kernel density, carrying the marginals.

Remark 4. *Every step of the two constructions is* mean-preserving*: barycentric atoms and mean-matched lognormal components hold the forward $\langle \eta, s\rangle = F$ exactly at each stage, so the approximation never leans on a renormalisation or a global rescaling and positivity is automatic. This is the population analogue of the per-fibre lock* (13)*, which fixes $\langle \mathcal{K}(s,\cdot), s'\rangle = s$ exactly at each regime—in both, the martingale identity is enforced by construction, not approached in the limit. So the ansatz is universal at both levels the construction needs—any marginal (Lemma 2) and any translation-invariant dynamics (Proposition 3)—with the leverage carrying the marginal match, so the kernel need only supply the dynamics.*

D Local identifiability of the dynamic kernel

The marginals do not pin the kernel, so it is fair to ask whether the generator $\boldsymbol{\theta}$ is identifiable from the data a desk has. Because every observable is closed-form in $\boldsymbol{\theta}$, this is a finite calculation. Form the elasticity-scaled Jacobian $\tilde{J}_{ij} = (\partial O_i/\partial \boldsymbol{\theta}_j)\,|\boldsymbol{\theta}_j|/|O_i|$ at the calibrated $\boldsymbol{\theta}^\star$, over nine observables (ATM vol @ 1m,1y; ATM skew @ 1m,1y; curvature @ 1m; SSR @ 1m,3m,6m,1y). Its singular values are

$$\sigma = (5.05,\, 2.96,\, 1.09,\, 0.59,\, 0.28,\, 0.19,\, \mathbf{0.059},\, \mathbf{0.011},\, \mathbf{0.000}). \tag{31}$$

Seven stiff directions, then a gap to two near-flat ones: **effective rank** $\approx 7/9$ at a 1% threshold (full algebraic rank, but condition number $\sim 10^4$). The softest singular vector loads on $\lambda_{\mathsf{S}}, \kappa_{\mathsf{S}}, \lambda_{\mathsf{F}}, \nu_{\mathsf{F}}$: the *SSR-shaping subspace.* Those four leverage/timescale knobs let the kernel fit the SSR, but they are not separately identified by a single date's smile and SSR term structure, because the SSR curve is too smooth to resolve the fast/slow split. (This is *not* the skew $\rho\xi$ pair, which here stays stiff.) The level, the vol-of-vol/curvature, the skew, and the SSR *level* are well-pinned; the fast-vs-slow SSR *decomposition* is the soft direction—the honest cost of the model's flexibility, and the price of a *right* SSR. (The spot-coupled six-knob variant was better-conditioned, cond $\approx$ 19, but only because its pinned long-end SSR was the wrong shape.)

Resolving the soft direction. To pin the fast/slow split, add a dynamic instrument the SSR term structure alone cannot carry. The closed-form VIX vol-of-vol term structure does most of it: VIXvov(τ), the stationary-π dispersion of the τ-forward vol, a finite sum over the regime grid. Adding VIXvov at 30 and 90 days to the nine observables drops the condition number from $\sim 10^4$ to ~ 840—a 12× regularisation, σ_9 up 15×. It resolves the *timescale* half (the κ, ν block) but is structurally blind to the leverages: forward-*variance* dispersion does not see λ. So the residual soft direction is the slow leverage λ_{S}, which forward-start skew—not vol-of-vol—must pin. The two dynamic instruments are complementary, not interchangeable.

Decoupling: the positive face. The same soft direction is the statics/dynamics *decoupling*. The directions that leave the smile fixed are $\ker J_{\text{stat}}$, and along them the SSR still moves, at the rate $\|P\,\nabla_{\boldsymbol{\theta}}\text{SSR}\|$ with $P = I - J_{\text{stat}}^{+} J_{\text{stat}}$ the static null-space projector. The resulting band at *fixed* statics ($\approx [0.93, 1.02]$) is reported in Section 7.1 (Figure 2).

Two limits. The statement is *local*, and it settles identifiability of the parameters *if* the market has this structure—not whether the kernel *fits* the market (fit quality, Section 7). Off-SPX, with no volatility-index options, the timescale split is separated only weakly—by the chain's own curvature and a realized estimate. So the (λ, κ) block is the fragile part when the readout is poor. Where data is thin, regularise it (a prior on the timescales) rather than floating all nine knobs.

E Closed-form ATM Taylor coefficients by implicit differentiation

This appendix derives the exact ATM implied-volatility coefficients used in Section 6.4 and writes them as explicit component sums; we verify them numerically at the end. (The same implicit-differentiation expansion appears in the GLIDE notes of Cao–Huang, unpublished.)

Let the conditional forward-return law be a Gaussian mixture $\mu = \sum_k \omega_k \mathcal{N}(m_k, v_k)$ in $z = \log(S_T/F)$, martingale-normalised so the forward is one: $\sum_k \omega_k F_k = 1$, $F_k = e^{m_k + \frac{1}{2}v_k}$. Work in log-strike $x = \log K$ and total volatility $u(x) = \widehat{\sigma}(x)\sqrt{T}$. The call price is the finite sum $C_{\text{mix}}(x) = \sum_k \omega_k \ \text{Call}(F_k, e^x, \sqrt{v_k})$ and the implied vol solves the price-matching identity

$$C_{\text{mix}}(x) = C_{\text{BS}}\big(x, u(x)\big), \qquad C_{\text{BS}}(x,u) = \Phi(d_+) - e^x\Phi(d_-), \quad d_\pm = -\frac{x}{u} \pm \frac{u}{2}. \tag{32}$$

Write $\delta_k := m_k/\sqrt{v_k}$ (the ATM d_- of component k), with φ, Φ the standard-normal density and CDF.

Level. At $x = 0$, $C_{\text{BS}}(0,u) = 2\Phi(u/2) - 1$, so the ATM total vol is one probit of the closed-form ATM mixture price:

$$u_0 = 2\,\Phi^{-1}\Big(\tfrac{1}{2} + \tfrac{1}{2}C_{\text{mix}}(0)\Big), \qquad C_{\text{mix}}(0) = \sum_k \omega_k \big[F_k\,\Phi(\delta_k + \sqrt{v_k}) - \Phi(\delta_k)\big]. \tag{33}$$

Skew and curvature. Differentiating (32) in x, with the total-vol Black greeks vega $\mathcal{V} = \partial_u C_{\text{BS}}$, vanna $\mathcal{V}_x = \partial^2_{xu} C_{\text{BS}}$, and volga $\mathcal{V}_u = \partial^2_u C_{\text{BS}}$, gives $\partial_x C_{\text{mix}} = \partial_x C_{\text{BS}} + \mathcal{V}\,u'$ and $\partial_x^2 C_{\text{mix}} = \partial_x^2 C_{\text{BS}} + 2\,\mathcal{V}_x\,u' + \mathcal{V}_u\,(u')^2 + \mathcal{V}\,u''$, which solve for $u' = u'(0)$ and $u'' = u''(0)$:

$$u' = \frac{\partial_x C_{\text{mix}} - \partial_x C_{\text{BS}}}{\mathcal{V}}, \qquad u'' = \frac{\partial_x^2 C_{\text{mix}} - \partial_x^2 C_{\text{BS}} - 2\,\mathcal{V}_x\,u' - \mathcal{V}_u\,(u')^2}{\mathcal{V}}. \tag{34}$$

Using the Black identity $F_k\varphi(\delta_k+\sqrt{v_k})=\varphi(\delta_k)$ to cancel the density terms, the mixture strike-derivatives at $x=0$ are the explicit sums

$$\partial_x C_{\text{mix}}(0)=-\sum_k \omega_k\,\Phi(\delta_k),\qquad \partial_x^2 C_{\text{mix}}(0)=\sum_k \omega_k\Big[\frac{\varphi(\delta_k)}{\sqrt{v_k}}-\Phi(\delta_k)\Big],\tag{35}$$

and the Black reference at $(x{=}0,u_0)$ is

$$\begin{gathered}\partial_x C_{\text{BS}}(0)=-\Phi\big(-\tfrac{u_0}{2}\big),\quad \partial_x^2 C_{\text{BS}}(0)=-\Phi\big(-\tfrac{u_0}{2}\big)+\frac{\varphi(u_0/2)}{u_0},\\ \mathcal{V}=\varphi\big(\tfrac{u_0}{2}\big),\quad \mathcal{V}_x=\tfrac{1}{2}\varphi\big(\tfrac{u_0}{2}\big),\quad \mathcal{V}_u=-\tfrac{u_0}{4}\varphi\big(\tfrac{u_0}{2}\big).\end{gathered}\tag{36}$$

Substituting the mixture sums (35) and the Black greeks (36) into (34) gives the total-vol skew $u_1=u'$ and curvature $u_2=u''$ in closed form; the annualised ATM coefficients are $\big(\widehat\sigma(0),\widehat\sigma'(0),\widehat\sigma''(0)\big)=(u_0,u_1,u_2)/\sqrt{T}$, with $\widehat\sigma'(0)=\mathrm{d}\widehat\sigma/\mathrm{d}\log K$ the SSR denominator (Section 6). Section 6.4 writes the same expansion in log-moneyness $m=-x$, where $\partial_m=-\partial_x$ flips the odd-order derivative.

Exactness (verified). Equations (33)–(36) are exact—no Black–Scholes inversion, only the probit Φ^{-1} for the level. On random martingale Gaussian mixtures they reproduce the bisection-inverted Black–Scholes implied vol and its finite-difference strike-derivatives to within $\sim 10^{-6}$ for the level and skew, and to finite-difference truncation for the curvature.

F The realised-SSR sampling band (HAC)

The realised SSR markers and the shaded band in Figures 4 (SPX) and 6 (NDX) are a physical-measure ($\mathbb{P}$) estimate from a single year of daily end-of-day chains, so both the numerator and the denominator carry sampling error that the band makes explicit. Comparing this $\mathbb{P}$ target to the kernel's risk-neutral SSR is legitimate at the daily horizon: the SSR is a ratio of one-step *covariations*, and quadratic covariation is measure-invariant—the $\mathbb{P}/\mathbb{Q}$ drift enters at $O(\mathrm{d}t)$ and drops out of the leading covariance—so the leverage reads the same under either measure to that order. The higher-moment vol-of-vol is *not* so protected (Section 7.2). At each tenor T_j ($j=1,\dots,5$; 1w to 3m) we regress the daily ATM-forward implied-vol change $\Delta\sigma^*_{j,t}$ on the daily log return r_t with an intercept, and divide the slope by the mean daily ATM skew $\bar{\mathcal{S}}_j=\overline{\partial_{\ln K}\sigma}\,|_{\text{ATM}}$ at T_j:

$$\Delta\sigma^*_{j,t}=a_j+\beta_j\,r_t+\varepsilon_{j,t},\qquad \widehat{R}_j\;=\;\beta_j/\bar{\mathcal{S}}_j.\tag{37}$$

The slope standard error is Newey–West with a Bartlett kernel (guaranteeing a non-negative variance) to absorb the serial correlation and heteroskedasticity of the daily residuals,

$$\widehat{\operatorname{Var}}(\hat\beta_j)=(X^\top X)^{-1}\Big(S_0+\sum_{\ell=1}^{L}\Big(1-\tfrac{\ell}{L+1}\Big)\big(S_\ell+S_\ell^\top\big)\Big)(X^\top X)^{-1},\qquad L=\big\lfloor 4\,(n/100)^{2/9}\big\rfloor,\tag{38}$$

where X is the design matrix, $S_\ell=\sum_t g_t\,g_{t-\ell}^\top$ the autocovariances of the scores $g_t=X_t\,\hat\varepsilon_{j,t}$, and the automatic lag is $L\approx 4$–5. Because R is a *ratio*, its error combines the slope and skew-denominator errors by the delta method,

$$\operatorname{se}(\widehat{R}_j)=\big|\widehat{R}_j\big|\sqrt{\Big(\tfrac{\operatorname{se}(\hat\beta_j)}{\beta_j}\Big)^2+\Big(\tfrac{\operatorname{se}(\bar{\mathcal{S}}_j)}{\bar{\mathcal{S}}_j}\Big)^2},\qquad \operatorname{se}(\bar{\mathcal{S}}_j)=\operatorname{std}(\mathcal{S}_{j,\cdot})/\sqrt{n},\tag{39}$$

and the plotted strip is $\widehat{R}_j \pm \mathrm{se}(\widehat{R}_j)$, one standard error per tenor. The scalar *noise floor* quoted in each panel title is this band as a single percentage—the tenor-RMS of the relative error,

$$\text{floor} \;=\; 100 \times \sqrt{\tfrac{1}{5}\sum_j \left(\mathrm{se}(\widehat{R}_j)/\widehat{R}_j\right)^2}\,\%. \tag{40}$$

The band is a *diagnostic*, not part of the objective: the kernel is fit to the point estimates $\widehat{R}_j$ in equal-percentage least squares (21), and the floor is the acceptance threshold—a fit whose SSR RMS falls below it is, on average, inside the $\pm$1-se strip and cannot be rejected against the realised target, so we stop there rather than chase sub-noise structure. The strip widens at the short end, where the slope regression has the fewest effective observations and the thinnest short-dated options. The forward-variance/VIX vol-of-vol readout (SPX, Figure 5) carries no band: it is a single-day risk-neutral ($\mathbb{Q}$) quote, not a sample statistic.

References

[1] Buehler, H., Horvath, B., Kratsios, A., Limmer, Y., Saqur, R. (2026). SANOS: Smooth strictly Arbitrage-free Non-parametric Option Surfaces. *arXiv preprint* arXiv:2601.11209.

[2] Dupire, B. (1994). Pricing with a smile. *Risk*, 7(1), 18–20.

[3] Breeden, D. T., Litzenberger, R. H. (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4), 621–651.

[4] Derman, E., Kani, I. (1994). Riding on a smile. *Risk*, 7(2), 32–39.

[5] Rubinstein, M. (1994). Implied binomial trees. *Journal of Finance*, 49(3), 771–818.

[6] Gyöngy, I. (1986). Mimicking the one-dimensional marginal distributions of processes having an Itô differential. *Probability Theory and Related Fields*, 71(4), 501–516.

[7] Jex, M., Henderson, R., Wang, D. (1999). Pricing exotics under the smile. *Risk*, 12(11), 72–75.

[8] Lipton, A. (2002). The vol smile problem. *Risk*, 15(2), 61–65.

[9] Heston, S.L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.

[10] Hagan, P.S., Kumar, D., Lesniewski, A.S., Woodward, D.E. (2002). Managing smile risk. *Wilmott Magazine*, September, 84–108.

[11] Corrado, C. J., Su, T. (1996). Skewness and kurtosis in S&P 500 index returns implied by option prices. *Journal of Financial Research*, 19(2), 175–192.

[12] Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley.

[13] Brigo, D., Mercurio, F. (2002). Lognormal-mixture dynamics and calibration to market volatility smiles. *International Journal of Theoretical and Applied Finance*, 5(4), 427–446.

[14] Ren, Y., Madan, D., Qian, M.Q. (2007). Calibrating and pricing with embedded local volatility models. *Risk*, 20(9), 138–143.

[15] Guyon, J., Henry-Labordère, P. (2012). Being particular about calibration. *Risk*, 25(1), 88–93.

[16] Guo, I., Loeper, G., Wang, S. (2022). Calibration of local-stochastic volatility models by optimal transport. *Mathematical Finance*, 32(1), 46–77. (First version: arXiv:1906.06478, 2019.)

[17] Cuchiero, C., Khosrawi, W., Teichmann, J. (2020). A generative adversarial network approach to calibration of local stochastic volatility models. *Risks*, 8(4), 101.

[18] Jourdain, B., Zhou, A. (2020). Existence of a calibrated regime switching local volatility model. *Mathematical Finance*, 30(2), 501–546. (arXiv:1607.00077.)

[19] Glasserman, P., Wu, Q. (2011). Forward and future implied volatility. *International Journal of Theoretical and Applied Finance*, 14(3), 407–432.

[20] Strassen, V. (1965). The existence of probability measures with given marginals. *Annals of Mathematical Statistics*, 36(2), 423–439.

[21] Beiglböck, M., Henry-Labordère, P., Penkner, F. (2013). Model-independent bounds for option prices—a mass transport approach. *Finance and Stochastics*, 17(3), 477–501.

[22] Bergomi, L. (2009). Smile dynamics IV. *Risk*, 22(12), 94–100.

[23] Bergomi, L. (2016). *Stochastic Volatility Modeling*. Chapman & Hall/CRC.

[24] Bergomi, L. (2017). Local-stochastic volatility: Models and non-models. *Risk*, August 2017, 78–83.

[25] Friz, P. K., Gatheral, J. (2024). Computing the SSR. *arXiv preprint* arXiv:2406.16131.

[26] Fukasawa, M. (2026). On the skew stickiness ratio. *arXiv preprint* arXiv:2602.05241.

[27] Bourgey, F., De Marco, S., Delemotte, J. (2024). Smile dynamics and rough volatility. *SSRN preprint* 4911186.

[28] Abi Jaber, E., Li, S. (2025). Capturing smile dynamics with the quintic volatility model: SPX, skew-stickiness ratio and VIX. *arXiv preprint* arXiv:2503.14158.

[29] Bandi, F. M., Fusari, N., Gazzani, G., Renò, R. (2026). Ultra-short-term volatility surfaces. *arXiv preprint* arXiv:2603.29430.

[30] van den Berg, T. (2026). Fast, reliable, and error-bounded option pricing with pretrained neural networks: A GJR–GARCH study. *arXiv preprint* arXiv:2606.15502.

[31] Guyon, J. (2022). Dispersion-constrained martingale Schrödinger bridges: Joint entropic calibration of stochastic volatility models to S&P 500 and VIX smiles. *SSRN preprint* 4165057.

[32] Che, C., Lin, H., Yang, Y., Hu, G., Fang, L. (2026). SPX–VIX risk computations via perturbed optimal transport. *arXiv preprint* arXiv:2603.10857.

[33] Doeff, E., Kamal, M. (2012). Consistent and realistic dynamics of the implied volatility surface under local volatility models. *SSRN preprint* 2207171.

[34] Hull, J., White, A. (2017). Optimal delta hedging for options. *Journal of Banking and Finance*, 82, 180–190.